

CCN PROJECT

TCP VS UDP TRAFFIC BEHAVIOR ANALYSIS FOR INTERNET APPLICATIONS USING WIRESHARK AND CISCO PACKET TRACER ON WINDOWS

REVIEW - 2

Team name: Protocol Pair





PROJECT STATEMENT

This project studies how TCP and UDP behave while handling traffic for common Internet applications using both Wireshark on a Windows system and Cisco Packet Tracer. Real-time network packets are captured and simulated to observe protocol behavior. Factors such as reliability, delay, packet loss, and throughput are analyzed, while simulation is used to visualize packet flow, including TCP's connection establishment and UDP's connectionless communication, to understand which protocol is better suited for different applications.

PROJECT DESCRIPTION

01 This is a fully software-based project carried out on a **Windows system**. **Wireshark** is used to observe and capture real-time network traffic from everyday Internet applications.



02 The project focuses on **understanding how TCP and UDP handle data differently during communication**.



03 Important performance factors such as **reliability, delay, packet loss, and throughput** are examined. The captured packet data is carefully analysed to **highlight the strengths and limitations of each protocol**.

04 This study helps in **identifying which protocol is better suited for specific Internet applications**.

TOOLS / PLATFORM USED



Operating System: Windows



Network Analysis Tool: Wireshark, Cisco packet tracer



Internet Applications: Web browser, file download, streaming services



Protocols Analyzed: TCP and UDP



Project Type: Completely software-based (no hardware involved)

LAYER-WISE NETWORKING

CONCEPTS APPLIED

01

Application Layer

- Generation of network traffic using common Internet applications
- Examples: Web browsing, file transfer, video streaming



02

Transport Layer

- Study of TCP and UDP protocols
- TCP concepts: Connection establishment, acknowledgements, retransmissions
- UDP concepts: Connectionless communication, faster data delivery



03

Network Layer

- IP addressing and packet routing
- Identification of source and destination IP addresses

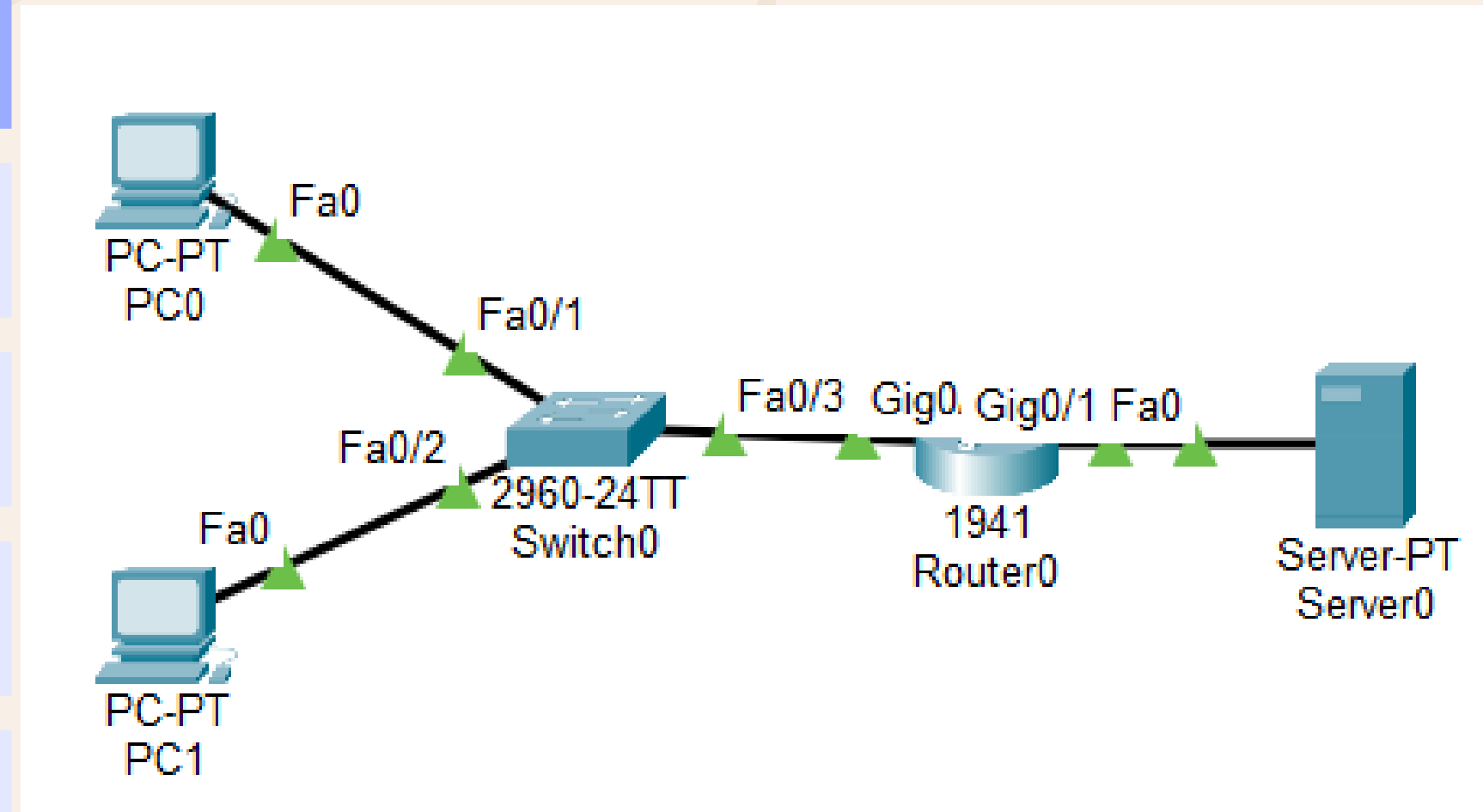
04

Data Link Layer

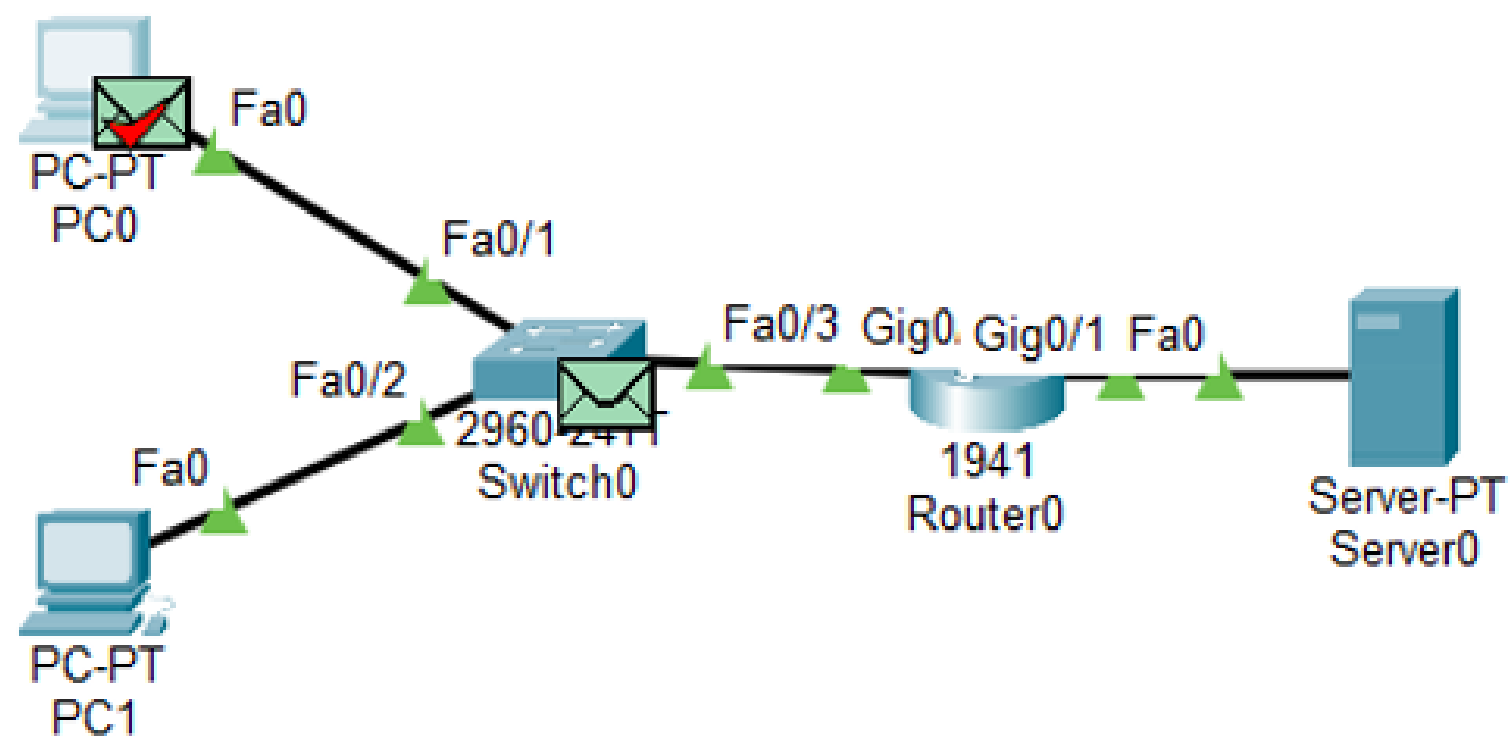
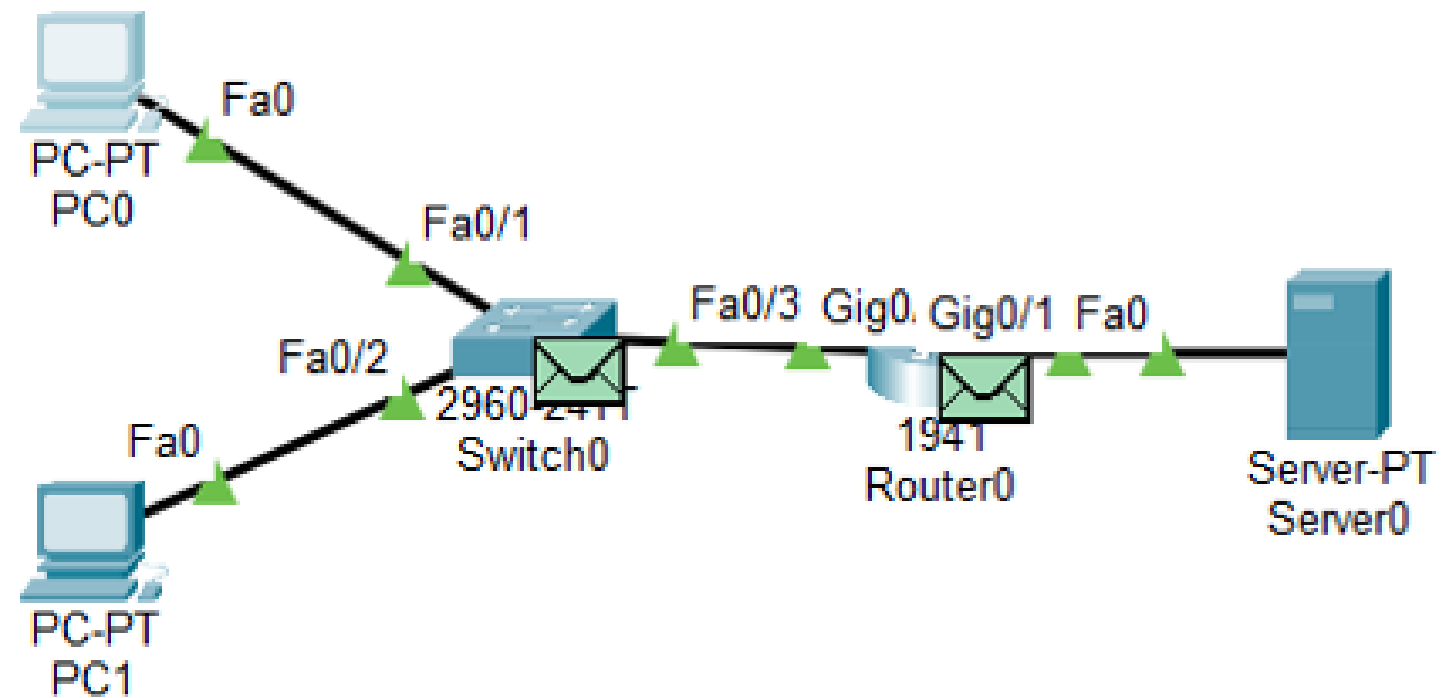
- Frame transmission over the local network
- MAC address identification during packet capture

DETAILS OF CISCO NETWORK

Device	Network Address	IPv4 Address	Gateway
PC0	192.168.1.0/24	192.168.1.2	192.168.1.1
PC1	192.168.1.0/24	192.168.1.3	192.168.1.1
Server0	192.168.2.0/24	192.168.2.10	192.168.2.1
Router G0/0	192.168.1.0/24	192.168.1.1	—
Router G0/1	192.168.2.0/24	192.168.2.1	—



PICTURES OF CISCO- TCP



Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.306	--	PC0	HTTP
	0.307	PC0	Switch0	TCP
	0.307	--	PC0	HTTP
	0.308	PC0	Switch0	HTTP
	0.308	Switch0	Router0	TCP
	0.309	Switch0	Router0	HTTP
	0.309	Router0	Server0	TCP
	0.310	Router0	Server0	HTTP
	0.311	Server0	Router0	HTTP
	0.312	Router0	Switch0	HTTP
	0.313	Switch0	PC0	HTTP
	0.313	--	PC0	TCP
	0.314	PC0	Switch0	TCP
	0.315	Switch0	Router0	TCP
	0.316	Router0	Server0	TCP
	0.317	Server0	Router0	TCP
	0.318	Router0	Switch0	TCP
	0.319	Switch0	PC0	TCP
	0.320	PC0	Switch0	TCP
	0.321	Switch0	Router0	TCP
	0.322	Router0	Server0	TCP

PDU Information at Device: Server0

OSI Model Inbound PDU Details

At Device: Server0
Source: PC0
Destination: 192.168.2.10

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer 4: TCP Src Port: 1026, Dst Port: 80	Layer4
Layer 3: IP Header Src. IP: 192.168.1.2, Dest. IP: 192.168.2.10	Layer3
Layer 2: Ethernet II Header 000B.BE7C.4E02 >> 0090.2BBB.BC75	Layer2
Layer 1: Port FastEthernet0	Layer1

PC0

Physical Config **Desktop** Programming Attributes

Web Browser X

< > URL http://192.168.2.10 Go Stop

Cisco Packet Tracer

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:
[A small page](#)
[Copyrights](#)
[Image page](#)
[Image](#)

PICTURES OF CISCO- UDP

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type **A Record**

Address

Add Save Remove

No.	Name	Type	Detail
0	example.com	A Record	192.168.2.10

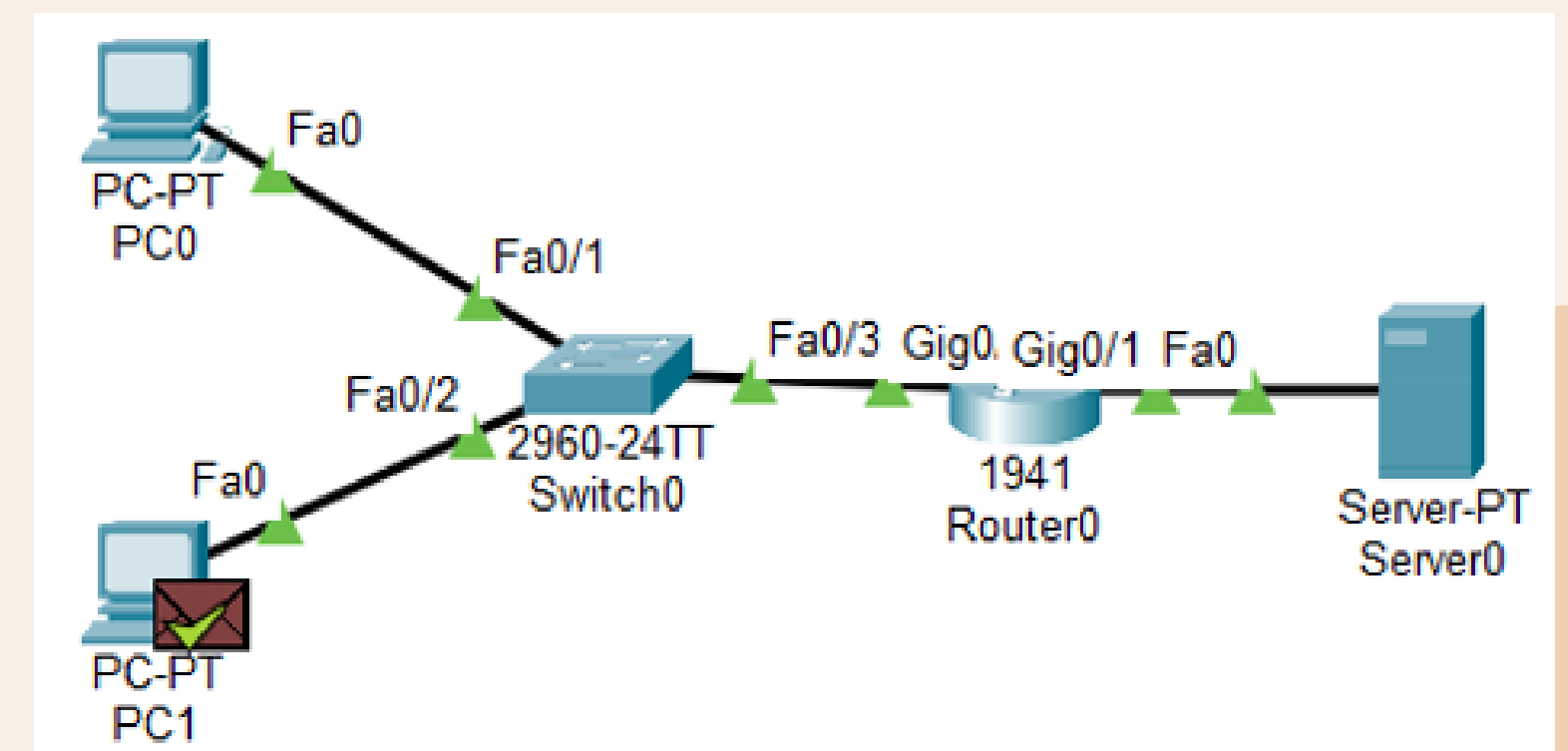
```
*** Request to 0.0.0.0 timed-out

C:\>nslookup example.com

Server: [192.168.2.10]
Address: 192.168.2.10

Non-authoritative answer:
Name: example.com
Address: 192.168.2.10

C:\>
```



PDU Information at Device: PC1

OSI Model

Outbound PDU Details

At Device: PC1
Source: PC1
Destination: 192.168.2.10

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer1

Out Layers

Layer 7: DNS

Layer6

Layer5

Layer 4: UDP Src Port: 1026, Dst Port: 53

Layer 3: IP Header Src. IP: 192.168.1.3, Dest. IP: 192.168.2.10

Layer 2: Ethernet II Header 0000.0C83.75E2 >> 000B.BE7C.4E01

Layer 1: Port(s): FastEthernet0

1. The device encapsulates the PDU into an UDP segment.

Simulation Panel					
Event List					
Vis.	Time(sec)	Last Device	At Device	Type	
	150.042	--	PC1		DNS
	150.043	PC1	Switch0		DNS
	150.044	Switch0	Router0		DNS
	150.045	Router0	Server0		DNS
	150.046	Server0	Router0		DNS
	150.047	Router0	Switch0		DNS
Visible	150.048	Switch0	PC1		DNS

WIRESHARK

UDP

udp						
No.	Time	Source	Destination	Protocol	Length	Info
10	0.504340	2406:7400:c4:c47f:c...	2001:4860:4860::8844	UDP	122	63549 → 443 Len=60
11	0.504729	2406:7400:c4:c47f:c...	2001:4860:4860::8844	UDP	122	63549 → 443 Len=60
12	0.505060	2406:7400:c4:c47f:c...	2001:4860:4860::8844	UDP	122	63549 → 443 Len=60
16	0.512122	2001:4860:4860::8844	2406:7400:c4:c47f:c...	UDP	97	443 → 63549 Len=31
17	0.512122	2001:4860:4860::8844	2406:7400:c4:c47f:c...	UDP	658	443 → 63549 Len=592
18	0.512122	2001:4860:4860::8844	2406:7400:c4:c47f:c...	UDP	575	443 → 63549 Len=509
19	0.512122	2001:4860:4860::8844	2406:7400:c4:c47f:c...	UDP	575	443 → 63549 Len=509
20	0.512122	2001:4860:4860::8844	2406:7400:c4:c47f:c...	UDP	89	443 → 63549 Len=23
21	0.512122	2001:4860:4860::8844	2406:7400:c4:c47f:c...	UDP	89	443 → 63549 Len=23
22	0.512122	2001:4860:4860::8844	2406:7400:c4:c47f:c...	UDP	89	443 → 63549 Len=23
23	0.513895	2406:7400:c4:c47f:c...	2001:4860:4860::8844	UDP	104	63549 → 443 Len=42
24	0.544647	2001:4860:4860::8844	2406:7400:c4:c47f:c...	UDP	91	443 → 63549 Len=25
52	4.523081	2406:7400:c4:c47f:c...	2001:4860:4860::8888	QUIC	1292	Initial, DCID=218d6a243545924b, PKN: 1, PING, CRYPTO, PADDING, CRYPTO, CRYPTO, CRYPTO, CRYPTO, PING, CRYPTO
53	4.523359	2406:7400:c4:c47f:c...	2001:4860:4860::8888	QUIC	1292	Initial, DCID=218d6a243545924b, PKN: 2, PING, CRYPTO, PING, PING, CRYPTO, CRYPTO, PING, CRYPTO
54	4.523925	2406:7400:c4:c47f:c...	2001:4860:4860::8888	QUIC	144	0-RTT, DCID=218d6a243545924b
55	4.524356	2406:7400:c4:c47f:c...	2001:4860:4860::8888	QUIC	324	0-RTT, DCID=218d6a243545924b
56	4.524532	2406:7400:c4:c47f:c...	2001:4860:4860::8888	QUIC	278	0-RTT, DCID=218d6a243545924b

> Frame 10: Packet, 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface \Device\NPF_{...}	0000	5c e9 31 b4 dc 2b 34 7d f6 af c3 46 86 dd 60 02	\.1...+4} ...F...`.
> Ethernet II, Src: Intel_af:c3:46 (34:7d:f6:af:c3:46), Dst: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b)	0010	63 c1 00 44 11 40 24 06 74 00 00 c4 c4 7f 0c 5b	c...D.@\$. t.....[
> Internet Protocol Version 6, Src: 2406:7400:c4:c47f:c5b:edec:f6f8:90e9, Dst: 2001:4860:4860::8844	0020	ed ec f6 f8 90 e9 20 01 48 60 48 60 00 00 00 00	 H`H`....
> User Datagram Protocol, Src Port: 63549, Dst Port: 443	0030	00 00 00 00 88 44 f8 3d 01 bb 00 44 18 d0 5a e4	D.= ...D..Z.
> Data (60 bytes)	0040	37 3d b7 9b 2f 02 21 51 2e 6c e0 7f dc 37 e8 34	7=.../·!Q .1...7·4
	0050	07 7d a9 d5 3b 70 6d d1 2e 97 87 5a 89 56 c0 13	·}...;pm· ...Z·V··
	0060	fe 1d 69 4c bc 9b 63 e1 75 f0 59 74 9e a5 31 1e	··iL··c· u·Yt··1·
	0070	06 16 05 9c ad 28 cc c5 ac 32	(·· ·2

Wireshark · Packet 19 · CCN_PROJECT_UDP.pcapng

> Frame 19: Packet, 575 bytes on wire (4600 bits), 575 bytes captured (4600 bit
> Ethernet II, Src: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b), Dst: Intel_af:c3:46 (3.
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
> Internet Protocol Version 6, Src: 2001:4860:4860::8844, Dst: 2406:7400:c4:c47
> User Datagram Protocol, Src Port: 443, Dst Port: 63549
> Data (509 bytes)

0000	34 7d f6 af c3 46 5c e9 31 b4 dc 2b 81 00 00 00	4}...F\ 1...+....
0010	86 dd 68 00 00 00 02 05 11 3b 20 01 48 60 48 60	..h..... ; .H`H`
0020	00 00 00 00 00 00 00 00 88 44 24 06 74 00 00 c4	 -D\$.t...
0030	c4 7f 0c 5b ed ec f6 f8 90 e9 01 bb f8 3d 02 05	...[..... =..
0040	87 ee 41 5d 82 c8 b6 bc 31 ac b8 2a a9 fb 22 f6	..A]..... 1..*..."
0050	48 ea dc cd 8d 6d 0a 84 95 78 20 ef c2 59 40 b3	H....m... -x ..Y@.
0060	87 8a 8a 8b 2f 3f 52 7c ca 33 07 96 1b ae c2 17	/?R -3.....
0070	b1 3e f1 3c 79 9c fd c5 e1 db 35 d4 ee 79 97 cd	..>.<y... ..5..y..
0080	73 ff 0f 2c fe 67 89 99 4d 31 fa a8 4b 2c c7 d9	s... ,.g... M1..K,..
0090	25 61 d6 69 36 3f 9d 8c 18 ea 8b ce 32 68 1f 83	%a.i6?...2h..
00a0	59 05 f7 9f b7 6b 12 c8 32 91 48 5a 9d 62 dd f7	Y....k... 2.HZ.b..

No.: 19 · Time: 0.512122 · Source: 2001:4860:4860::8844 · Destin...8:90e9 · Protocol: UDP · Length: 575 · Info: 443 → 63549 Len=509

☒ Show packet bytes Layout: Vertical (Stacked) Close Help

UDP

- Connectionless
- No sequence tracking (at transport level)
- No ACK
- No flow control

UDP

Wireshark · Conversations · CCN_PROJECT_UDP.pcapng

Conversation Settings

☐ Name resolution

☐ Absolute start time

☐ Display raw data

☒ Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

Protocol

☐ Bluetooth

☐ BPv7

☐ DCCP

☐ DNP 3.0

☒ Ethernet

☐ FC

☐ FDDI

☐ IEEE 802.11

☐ IEEE 802.15.4

☐ ICMP

Filter list for specific type

Ethernet · 17IPv4 · 35IPv6 · 115TCPUDP · 1044

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel St
0.0.0.0	68	255.255.255.255	67	2	954 bytes	636	2	100.00%	2	954 bytes	0	0 bytes	2658.6206
192.168.0.1	67	192.168.0.109	68	1	590 bytes	790	1	100.00%	1	590 bytes	0	0 bytes	3515.9858
192.168.0.1	1900	192.168.0.109	61437	3	2 kB	691	3	100.00%	3	2 kB	0	0 bytes	2742.3057
192.168.0.1	1900	192.168.0.109	59369	2	1 kB	805	2	100.00%	2	1 kB	0	0 bytes	3516.9496
192.168.0.1	1900	192.168.0.109	58798	14	7 kB	823	14	100.00%	14	7 kB	0	0 bytes	3517.7565
192.168.0.1	57604	239.255.255.250	1900	523	263 kB	13	523	100.00%	523	263 kB	0	0 bytes	22.94811
192.168.0.101	33109	192.168.0.109	51096	1	545 bytes	842	1	100.00%	1	545 bytes	0	0 bytes	3519.4269
192.168.0.101	33666	192.168.0.109	51096	1	545 bytes	889	1	100.00%	1	545 bytes	0	0 bytes	3521.5441
192.168.0.101	45528	192.168.0.109	51096	1	545 bytes	908	1	100.00%	1	545 bytes	0	0 bytes	3524.2213
192.168.0.101	48550	192.168.0.109	51096	1	545 bytes	880	1	100.00%	1	545 bytes	0	0 bytes	3520.9716
192.168.0.101	48826	192.168.0.109	51096	1	545 bytes	843	1	100.00%	1	545 bytes	0	0 bytes	3519.4449
192.168.0.101	53340	192.168.0.109	51096	1	545 bytes	876	1	100.00%	1	545 bytes	0	0 bytes	3520.9504
192.168.0.101	59652	192.168.0.109	51096	1	545 bytes	859	1	100.00%	1	545 bytes	0	0 bytes	3520.4514
192.168.0.101	48550	255.255.255.255	51096	1	654 bytes	883	1	100.00%	1	654 bytes	0	0 bytes	3520.9955
192.168.0.101	53340	255.255.255.255	51096	1	654 bytes	879	1	100.00%	1	654 bytes	0	0 bytes	3520.9716
192.168.0.106	33187	192.168.0.109	58798	1	455 bytes	904	1	100.00%	1	455 bytes	0	0 bytes	3523.2938
192.168.0.106	34370	192.168.0.109	58798	1	365 bytes	907	1	100.00%	1	365 bytes	0	0 bytes	3523.8273
192.168.0.106	34429	192.168.0.109	59369	1	368 bytes	808	1	100.00%	1	368 bytes	0	0 bytes	3517.0632
192.168.0.106	36891	192.168.0.109	58798	1	455 bytes	919	1	100.00%	1	455 bytes	0	0 bytes	3526.3846
192.168.0.106	39057	192.168.0.109	58798	1	663 bytes	918	1	100.00%	1	663 bytes	0	0 bytes	3526.3249
192.168.0.106	39084	192.168.0.109	58798	1	359 bytes	906	1	100.00%	1	359 bytes	0	0 bytes	3523.8273
192.168.0.106	39222	192.168.0.109	58798	1	455 bytes	899	1	100.00%	1	455 bytes	0	0 bytes	3523.2044
192.168.0.106	39535	192.168.0.109	58798	1	663 bytes	888	1	100.00%	1	663 bytes	0	0 bytes	3521.3623
192.168.0.106	39984	192.168.0.109	58798	1	663 bytes	835	1	100.00%	1	663 bytes	0	0 bytes	3518.3657
192.168.0.106	43557	192.168.0.109	58798	1	365 bytes	896	1	100.00%	1	365 bytes	0	0 bytes	3523.1996
192.168.0.106	44231	192.168.0.109	59369	1	455 bytes	807	1	100.00%	1	455 bytes	0	0 bytes	3516.9857

Close

Help

TCP:

tcp

No.	Time	Source	Destination	Protocol	Length	Info
4	0.149193	2406:7400:c4:c47f:c...	2600:140f:3::6856:b...	TLSv1.2	1313	Application Data
5	0.149304	2406:7400:c4:c47f:c...	2600:140f:3::6856:b...	TLSv1.2	105	Application Data
6	0.156016	2600:140f:3::6856:b...	2406:7400:c4:c47f:c...	TCP	74	443 → 56567 [ACK] Seq=1 Ack=1322 Win=586 Len=0
7	0.266063	2600:140f:3::6856:b...	2406:7400:c4:c47f:c...	TLSv1.2	196	Application Data
8	0.266063	2600:140f:3::6856:b...	2406:7400:c4:c47f:c...	TLSv1.2	105	Application Data
9	0.266146	2406:7400:c4:c47f:c...	2600:140f:3::6856:b...	TCP	74	56567 → 443 [ACK] Seq=1322 Ack=154 Win=1020 Len=0
13	0.505343	2406:7400:c4:c47f:c...	2603:1063:2200:8::20	TLSv1.2	218	Application Data
14	0.505587	2406:7400:c4:c47f:c...	2603:1063:2200:8::20	TLSv1.2	113	Application Data
15	0.505676	2406:7400:c4:c47f:c...	2603:1063:2200:8::20	TLSv1.2	13764	Application Data
26	0.747059	2603:1063:2200:8::20	2406:7400:c4:c47f:c...	TCP	74	443 → 54638 [ACK] Seq=1 Ack=184 Win=16380 Len=0
27	0.747059	2603:1063:2200:8::20	2406:7400:c4:c47f:c...	TLSv1.2	113	Application Data
28	0.747059	2603:1063:2200:8::20	2406:7400:c4:c47f:c...	TCP	74	443 → 54638 [ACK] Seq=40 Ack=1584 Win=16385 Len=0
29	0.747059	2603:1063:2200:8::20	2406:7400:c4:c47f:c...	TCP	74	443 → 54638 [ACK] Seq=40 Ack=2984 Win=16380 Len=0
30	0.747059	2603:1063:2200:8::20	2406:7400:c4:c47f:c...	TCP	74	443 → 54638 [ACK] Seq=40 Ack=4384 Win=16385 Len=0
31	0.749857	2603:1063:2200:8::20	2406:7400:c4:c47f:c...	TCP	74	443 → 54638 [ACK] Seq=40 Ack=5784 Win=16380 Len=0
32	0.749857	2603:1063:2200:8::20	2406:7400:c4:c47f:c...	TCP	74	443 → 54638 [ACK] Seq=40 Ack=7184 Win=16385 Len=0
33	0.751855	2603:1063:2200:8::20	2406:7400:c4:c47f:c...	TCP	74	443 → 54638 [ACK] Seq=40 Ack=8584 Win=16380 Len=0

> Frame 15: Packet, 13764 bytes on wire (110112 bits), 13764 bytes captured (110112 bits) on interface \Device\NPF{...}

> Ethernet II, Src: Intel_af:c3:46 (34:7d:f6:af:c3:46), Dst: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b)

> Internet Protocol Version 6, Src: 2406:7400:c4:c47f:c5b:edec:f6f8:90e9, Dst: 2603:1063:2200:8::20

> Transmission Control Protocol, Src Port: 54638, Dst Port: 443, Seq: 184, Ack: 1, Len: 13690

> Transport Layer Security

0000 5c e9 31 b4 dc 2b 34 7d f6 af c3 46 86 dd 60 0b \.1...+4} ...F...`

0010 a2 58 35 8e 06 40 24 06 74 00 00 c4 c4 7f 0c 5b .X5...@\$...t.....[

0020 ed ec f6 f8 90 e9 26 03 10 63 22 00 00 08 00 00&...c".....

0030 00 00 00 00 00 20 d5 6e 01 bb fc eb 98 06 56 a8n.....V.

0040 44 6f 50 18 00 fc 38 09 00 00 17 03 03 35 75 50 DoP...8...5uP

0050 30 d2 da 18 7b ed 45 5b 97 a8 14 e8 f9 ed 08 d2 0...{.E[.....

0060 a6 ca 35 3a 41 d5 3a 3f 14 74 17 7c f7 ad b9 bd ..5:A.:? .t.|....

0070 3b ce 13 09 4a 35 ff bf d4 12 d8 29 da 80 c8 e0 ;...J5... ..)....

0080 1f 08 18 05 8a 17 d1 cd da e3 c1 33 fe 8a 76 f53...v.

0090 92 16 83 dd 8e 3b f4 29 ba a2 aa ca cc aa f1 ee;...).....

00a0 eb a7 d0 d8 34 d5 fe 2b 4d d5 ca 7a 88 99 7e 474...+ M...z...~G

00b0 d3 61 be fa 21 45 34 ff 14 98 e3 c6 f5 1c ce 35 .a...!E4... ..5

00c0 c8 30 2f 8e be 26 73 00 ab 09 9d 3a 86 8f fe 32 .0/...&s... ..:...2

00d0 87 6e 23 7e 72 da 20 4f 7b 19 39 52 2a 1f 9c db .n#~r... 0 {..9R*...

00e0 ea c4 53 30 2f ab ec cb 67 9b ff 2e a4 a9 82 d6 ..S0/... g... ..

00f0 32 88 ce ff 1e 17 40 88 ba f2 51 4c d1 df 5e 00 2...@... ..QL...^.

0100 54 9e cd 57 dc 3c 54 8b 4e a7 f6 01 65 a3 d1 c0 T..W.<T.. N...e...

0110 4b ba ce 48 3e 2a 2f 8d c7 12 a6 8a bc 11 fe e2 K..H>*/... ..

0120 5a c9 51 c0 59 ef 94 28 1d 2e dc 01 d1 79 51 08 Z.Q.Y..(...yQ.

0130 f1 de c3 2d fc c9 15 92 11 c3 5b 55 97 90 53 30[U..S0

0140 9b 11 ee f2 90 4e 56 4f dd 2b d9 89 83 59 9f 4aNVO ..+...Y..J

0150 89 a3 e3 95 4a 58 a2 2e ed 50 70 64 a1 e7 9f a5JX... .Ppd....

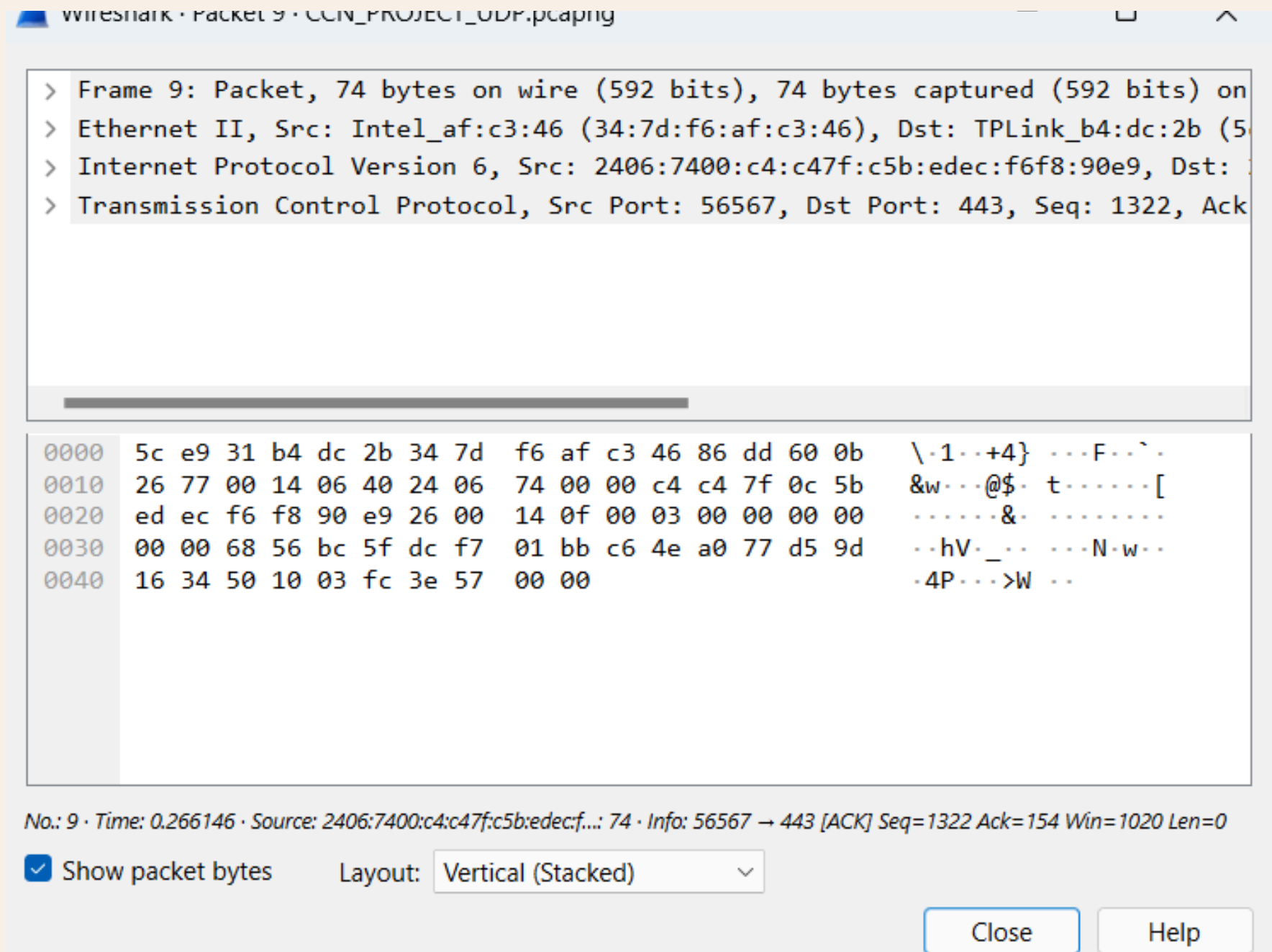
0160 bb 62 fc 47 ba 8f 00 b4 14 c4 71 4e 26 d5 d3 e1 .b.G.... ..qN&...

Transmission Control Protocol: Protocol

Packets: 127261 · Displayed: 67884 (53.3%)

Profile: Default

1 packet TCP:



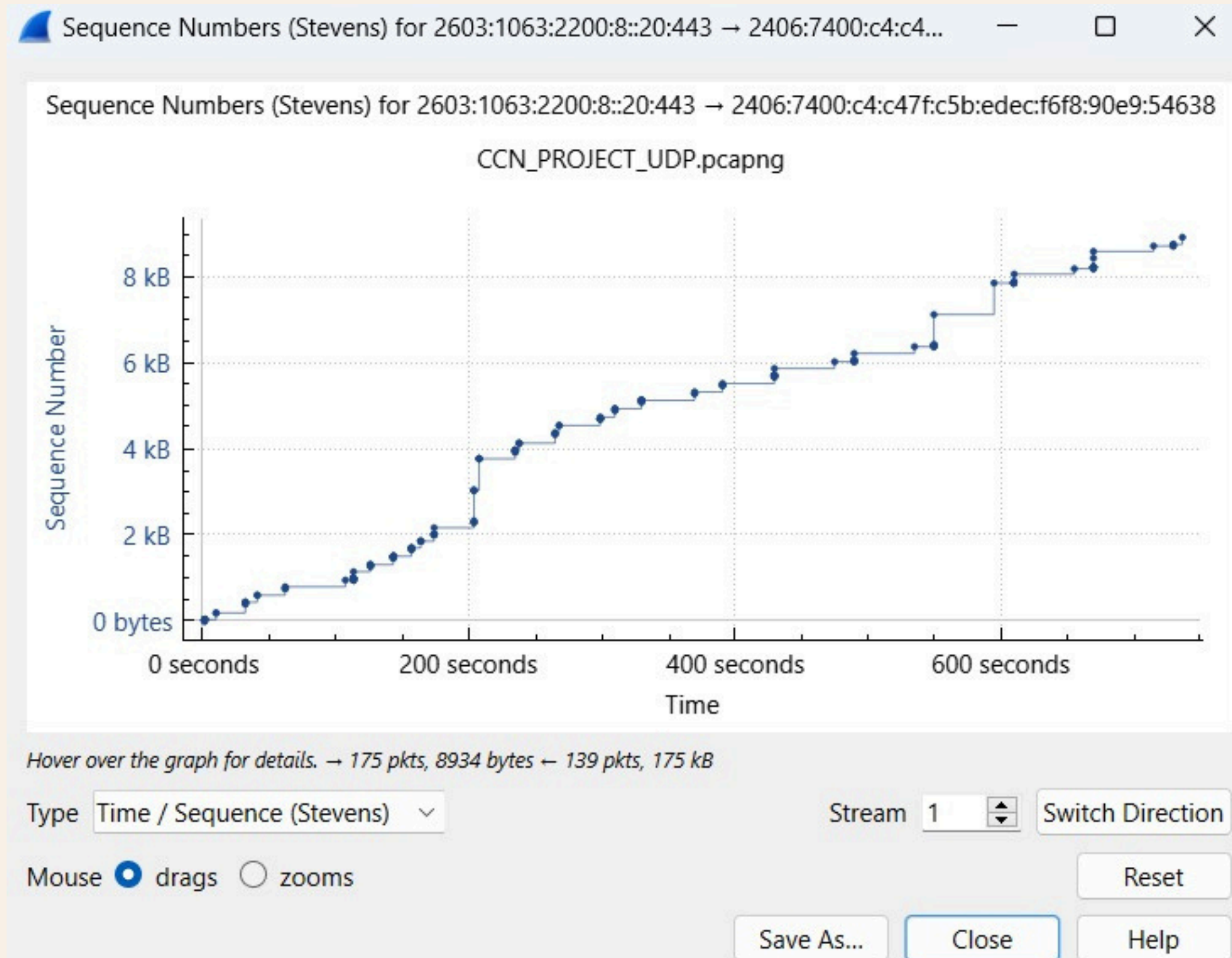
TCP

- Connection-oriented
- Has sequence numbers
- Has acknowledgments
- Tracks data flow

SYN , SYN ACK:

tcp.flags.syn == 1						
No.	Time	Source	Destination	Protocol	Length	Info
371	8.644065	2406:7400:c4:c47f:c...	2600:140f:3::685d:1...	TCP	86	52892 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 WS=256 SACK_PERM
374	8.647186	2600:140f:3::685d:1...	2406:7400:c4:c47f:c...	TCP	86	443 → 52892 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1344 SACK_PERM WS=128
919	29.138149	192.168.0.109	20.189.173.13	TCP	66	52893 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
920	29.341036	20.189.173.13	192.168.0.109	TCP	66	443 → 52893 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
927	29.478032	2406:7400:c4:c47f:c...	2603:1063:5:122::36...	TCP	86	52894 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 WS=256 SACK_PERM
928	29.481301	2603:1063:5:122::36...	2406:7400:c4:c47f:c...	TCP	86	443 → 52894 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
942	29.573748	2406:7400:c4:c47f:c...	2603:1063:5:122::36...	TCP	86	52895 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 WS=256 SACK_PERM
943	29.579882	2603:1063:5:122::36...	2406:7400:c4:c47f:c...	TCP	86	443 → 52895 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
978	30.405765	192.168.0.109	20.189.173.13	TCP	66	52896 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
981	30.443658	2406:7400:c4:c47f:c...	2603:1063:5:122::36...	TCP	86	52897 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 WS=256 SACK_PERM
982	30.444490	2406:7400:c4:c47f:c...	2603:1063:5:122::36...	TCP	86	52898 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 WS=256 SACK_PERM
984	30.446637	2603:1063:5:122::36...	2406:7400:c4:c47f:c...	TCP	86	443 → 52897 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
986	30.446987	2603:1063:5:122::36...	2406:7400:c4:c47f:c...	TCP	86	443 → 52898 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
1004	30.608563	20.189.173.13	192.168.0.109	TCP	66	443 → 52896 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1023	30.939077	2406:7400:c4:c47f:c...	2603:1063:2001:1d2b...	TCP	86	52899 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 WS=256 SACK_PERM
1024	30.945409	2603:1063:2001:1d2b...	2406:7400:c4:c47f:c...	TCP	86	443 → 52899 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
12805	68.646328	2406:7400:c4:c47f:c...	2600:140f:3::685d:1...	TCP	86	51649 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 WS=256 SACK_PERM
> Frame 371: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{A...					0000	5c e9 31 b4 dc 2b 34 7d f6 af c3 46 86 dd 60 05 \.1..+4} ...F..`.
> Ethernet II, Src: Intel_af:c3:46 (34:7d:f6:af:c3:46), Dst: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b)					0010	c0 3f 00 20 06 40 24 06 74 00 00 c4 c4 7f 0c 5b .?. @\$ t.....[
> Internet Protocol Version 6, Src: 2406:7400:c4:c47f:c5b:edec:f6f8:90e9, Dst: 2600:140f:3::685d:19a2					0020	ed ec f6 f8 90 e9 26 00 14 0f 00 03 00 00 00 00&.....
> Transmission Control Protocol, Src Port: 52892, Dst Port: 443, Seq: 0, Len: 0					0030	00 00 68 5d 19 a2 ce 9c 01 bb 8d 89 fe 08 00 00 ..h]....
					0040	00 00 80 02 fd 20 9b ac 00 00 02 04 05 a0 01 03
					0050	03 08 01 01 04 02

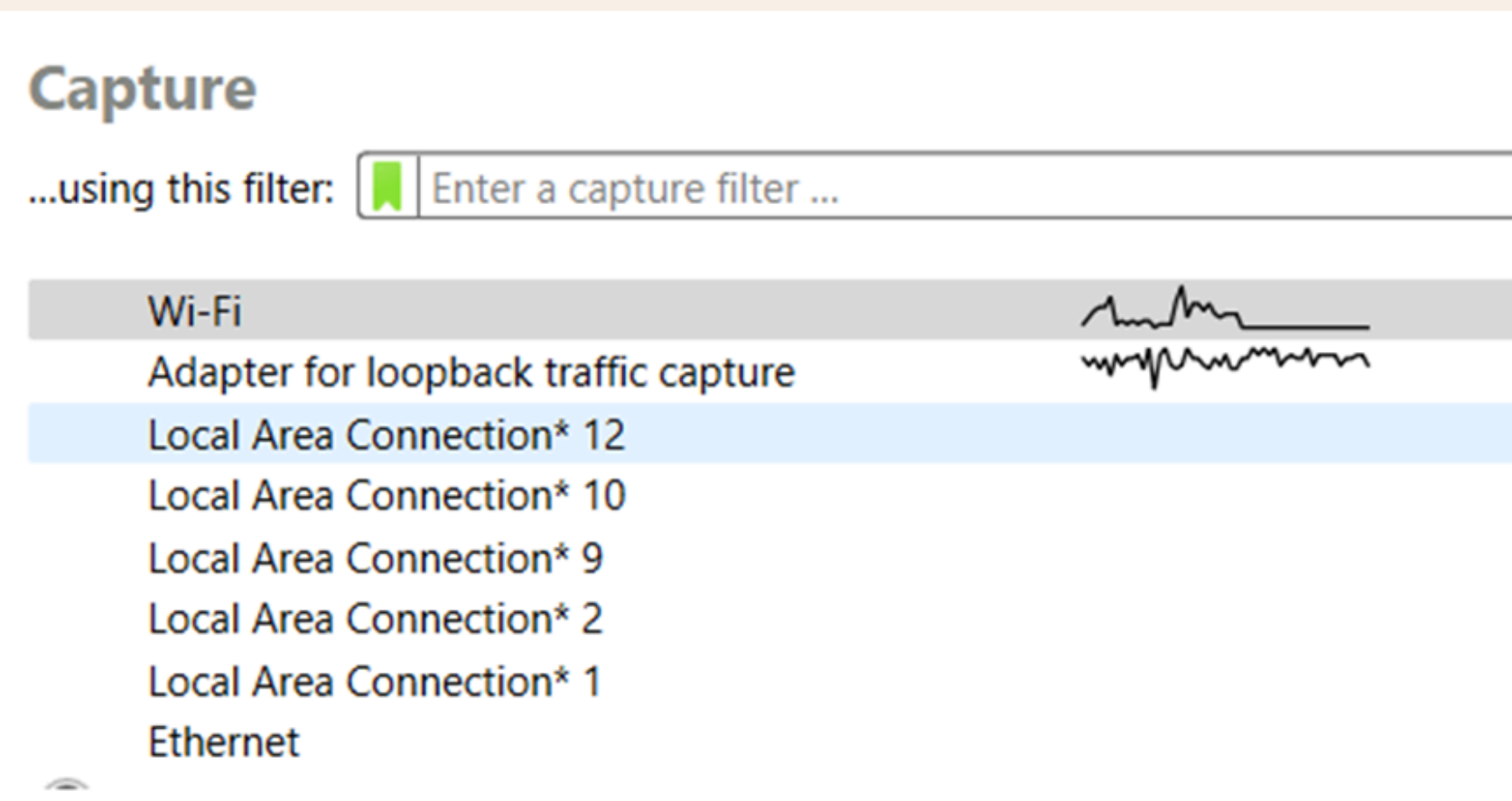
TCP - Time Sequence Graph:



Feature	TCP	UDP
Handshake	Seen	Not seen
ACK	Present	Not present
Graph	Available	Not available
Packet flow	Controlled	Not available
Conversations	Yes	Not available

TCP provides reliable, connection-oriented communication with error control and ordered delivery, whereas UDP provides fast, connectionless communication without reliability, making it suitable for real-time applications.

Selection of which network to be captured:



The graph near the Wi-Fi etc., show that shows that the traffic exists.

- Spikes indicate packets being transmitted
- Flat line indiactes no traffic

Apply a display filter ... <Ctrl-/>							
No.	Time	Source	Destination	Protocol	Length	Info	
649	4.266228	192.168.0.102	170.114.117.9	UDP	69	54712 → 8801	Len=27
650	4.268176	170.114.117.9	192.168.0.102	UDP	1115	8801 → 54712	Len=1069
651	4.268176	170.114.117.9	192.168.0.102	UDP	1115	8801 → 54712	Len=1069
652	4.272032	170.114.117.9	192.168.0.102	UDP	340	8801 → 60998	Len=294
653	4.272032	170.114.117.9	192.168.0.102	UDP	1115	8801 → 54712	Len=1069
654	4.304916	170.114.117.9	192.168.0.102	UDP	1115	8801 → 54712	Len=1069
655	4.304916	170.114.117.9	192.168.0.102	UDP	321	8801 → 54712	Len=275
656	4.304916	170.114.117.9	192.168.0.102	UDP	347	8801 → 60998	Len=301
657	4.311136	170.114.117.9	192.168.0.102	UDP	356	8801 → 60998	Len=310
658	4.311507	192.168.0.102	170.114.117.9	UDP	69	54712 → 8801	Len=27
659	4.337478	170.114.117.9	192.168.0.102	UDP	339	8801 → 60998	Len=293
660	4.359989	170.114.117.9	192.168.0.102	UDP	334	8801 → 60998	Len=288
661	4.359989	170.114.117.9	192.168.0.102	UDP	905	8801 → 54712	Len=859
662	4.359989	170.114.117.9	192.168.0.102	UDP	905	8801 → 54712	Len=859
663	4.359989	170.114.117.9	192.168.0.102	UDP	905	8801 → 54712	Len=859
664	4.371015	192.168.0.102	170.114.117.9	UDP	76	54712 → 8801	Len=34

This picture shows that the packets are flowing.

Filtering only TCP in the flowing packets

- Only TCP packets are displayed
- Removes noise from UDP/ICMP

To show only tcp, source port,
destination port etc.,

Filter used: `tcp.flags.syn == 1`

tcp						
No.	Time	Source	Destination	Protocol	Length	Info
15	0.095764	192.168.0.102	148.113.20.113	TCP	55	49602 → 80 [ACK] Seq=1 Ack=1 Win=516 Len=1
20	0.130757	148.113.20.113	192.168.0.102	TCP	70	80 → 49602 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
76	0.549702	170.114.117.9	192.168.0.102	TLSv1.2	137	Application Data
77	0.549856	192.168.0.102	170.114.117.9	TLSv1.2	104	Application Data
83	0.610078	170.114.117.9	192.168.0.102	TCP	58	443 → 58714 [ACK] Seq=80 Ack=51 Win=16 Len=0
109	0.808371	170.114.117.9	192.168.0.102	TLSv1.2	130	Application Data
110	0.809152	192.168.0.102	170.114.117.9	TLSv1.2	97	Application Data
113	0.834104	170.114.117.9	192.168.0.102	TCP	58	443 → 58714 [ACK] Seq=152 Ack=94 Win=16 Len=0
142	1.047619	2406:7400:c4:f8b7:a...	2a03:2880:f335:120:...	TLSv1.2	16480	Application Data
143	1.047764	2406:7400:c4:f8b7:a...	2a03:2880:f335:120:...	TLSv1.2	2413	Application Data
146	1.059518	2a03:2880:f335:120:...	2406:7400:c4:f8b7:a...	TCP	74	443 → 50928 [ACK] Seq=1 Ack=1393 Win=5399 Len=0
147	1.059518	2a03:2880:f335:120:...	2406:7400:c4:f8b7:a...	TCP	74	443 → 50928 [ACK] Seq=1 Ack=2785 Win=5399 Len=0
148	1.059518	2a03:2880:f335:120:...	2406:7400:c4:f8b7:a...	TCP	74	443 → 50928 [ACK] Seq=1 Ack=4177 Win=5399 Len=0
149	1.059518	2a03:2880:f335:120:...	2406:7400:c4:f8b7:a...	TCP	74	443 → 50928 [ACK] Seq=1 Ack=5569 Win=5399 Len=0
150	1.059518	2a03:2880:f335:120:...	2406:7400:c4:f8b7:a...	TCP	74	443 → 50928 [ACK] Seq=1 Ack=6961 Win=5399 Len=0
151	1.059518	2a03:2880:f335:120:...	2406:7400:c4:f8b7:a...	TCP	74	443 → 50928 [ACK] Seq=1 Ack=8353 Win=5399 Len=0
152	1.059518	2a03:2880:f335:120:...	2406:7400:c4:f8b7:a...	TCP	74	443 → 50928 [ACK] Seq=1 Ack=9745 Win=5399 Len=0

> Frame 15: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{A...}

> Ethernet II, Src: Intel_af:c3:46 (34:7d:f6:af:c3:46), Dst: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b)

> Internet Protocol Version 4, Src: 192.168.0.102, Dst: 148.113.20.113

> Transmission Control Protocol, Src Port: 49602, Dst Port: 80, Seq: 1, Ack: 1, Len: 1

0000

5c e9 31 b4 dc 2b 34 7d f6 af c3 46 08 00 45 00

\.1...+4} ...F..E.

0010

00 29 10 05 40 00 80 06 00 00 c0 a8 00 66 94 71

..).@... ..f.q

0020

14 71 c1 c2 00 50 3d 9a 42 fe 87 33 70 35 50 10

..q...P= B..3p5P

0030

02 04 6a 0c 00 00 00

..j....

```
Transmission Control Protocol, Src Port: 443, Dst Port: 50928, Seq: 1, Ack: 1393, Len: 0
  Source Port: 443
  Destination Port: 50928
  [Stream index: 2]
  [Stream Packet Number: 3]
  > [Conversation completeness: Incomplete (12)]
  [TCP Segment Len: 0]
```

```
> [Conversation completeness: Incomplete (12)]
  [TCP Segment Len: 0]
  Sequence Number: 1      (relative sequence number)
  Sequence Number (raw): 1192021852
  [Next Sequence Number: 1      (relative sequence number)]
  Acknowledgment Number: 1393      (relative ack number)
  Acknowledgment number (raw): 1056109085
  0101 ..... = Header Length: 20 bytes (5)
```

3 way handshake

Connection establishment process:

1.SYN

- Client to Server
- Requests connection

2.SYN-ACK

- Server to Client
- Accepts request

3.ACK

- Client to Server
- Confirms connection

After opening a https website:

TCP for the syn in the 3-way handshake:

*Wi-Fi

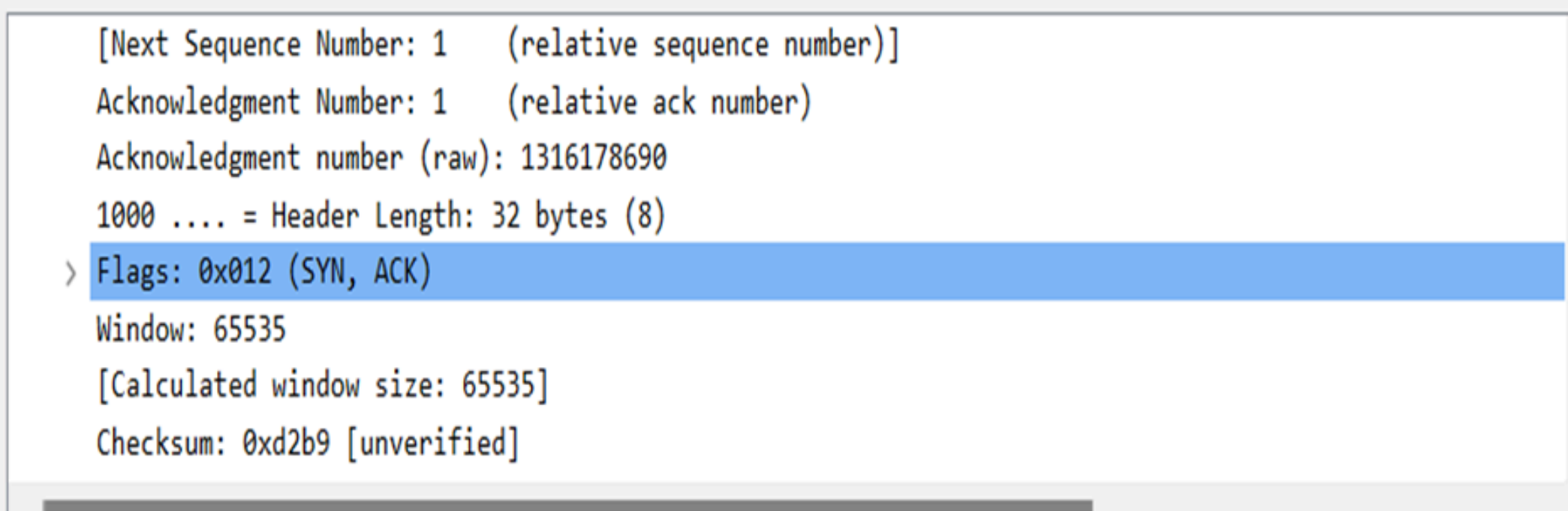
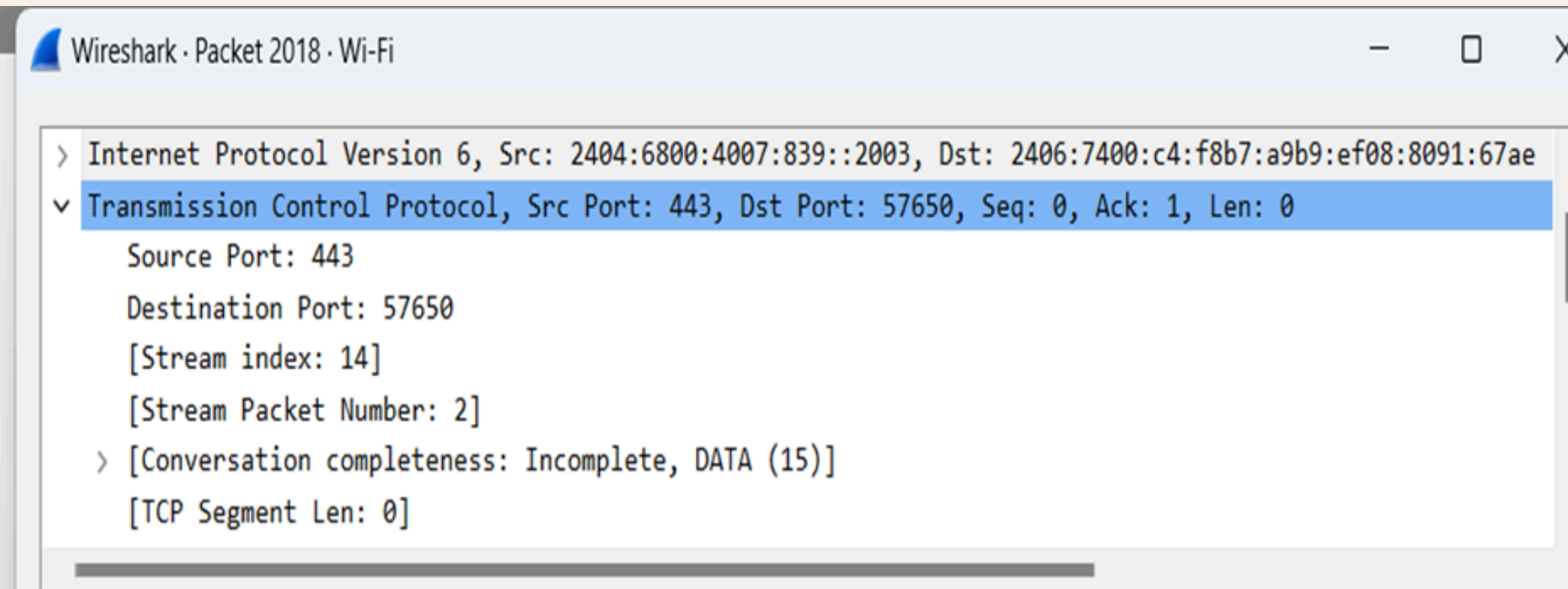
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn == 1

No.	Time	Source	Destination	Protocol	Length	Info
2016	12.980547	2406:7400:c4:f8b7:a...	2404:6800:4007:839:...	TCP	86	57650 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 WS=256 SACK_PERM
2018	12.982388	2404:6800:4007:839:...	2406:7400:c4:f8b7:a...	TCP	90	443 → 57650 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1420 SACK_PERM WS=256

```
> Ethernet II, Src: Intel_af:c3:46 (34:7d:f6:af:c3:46), Dst: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b)
> Internet Protocol Version 6, Src: 2406:7400:c4:f8b7:a9b9:ef08:8091:67ae, Dst: 2404:6800:4007:839::200
v Transmission Control Protocol, Src Port: 57650, Dst Port: 443, Seq: 0, Len: 0
  Source Port: 57650
  Destination Port: 443
  [Stream index: 14]
  [Stream Packet Number: 1]
  > [Conversation completeness: Incomplete, DATA (15)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 1316178689
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x002 (SYN)
  Window: 64800
  [Calculated window size: 64800]
```

TCP for the syn-ack :



Flags control communication

- SYN indicates start
- ACK indicates acknowledgment

TCP uses flags to manage connection states and ensure proper communication.

TCP RETRANSMISSION:

tcp.analysis.retransmission						
No.	Time	Source	Destination	Protocol	Length	Info
535	13.333929	2406:7400:c4:c47f:c...	2603:1063:2000:1::12	TCP	1474	[TCP Retransmission] 60179 → 443 [PSH, ACK] Seq=6331 Ack=1 Win=510 Len=1400
577	16.153675	192.168.0.109	20.189.173.17	TCP	89	[TCP Retransmission] 57782 → 443 [PSH, ACK] Seq=5644 Ack=153 Win=255 Len=35
13822	84.682897	2603:1063:2000:1::12	2406:7400:c4:c47f:c...	TCP	570	[TCP Retransmission] 443 → 60179 [PSH, ACK] Seq=4854 Ack=48136 Win=49152 Len=496
13829	84.682897	2603:1063:2000:1::12	2406:7400:c4:c47f:c...	TCP	570	[TCP Retransmission] 443 → 60179 [PSH, ACK] Seq=4854 Ack=48136 Win=49152 Len=496
15344	112.526535	2406:7400:c4:c47f:c...	2603:1063:2000:1::12	TCP	1474	[TCP Retransmission] 60179 → 443 [PSH, ACK] Seq=61809 Ack=5811 Win=510 Len=1400
15490	112.753380	2603:1063:2000:1::12	2406:7400:c4:c47f:c...	TCP	557	[TCP Spurious Retransmission] 443 → 60179 [PSH, ACK] Seq=5811 Ack=63209 Win=49152 Len=483
16294	122.206609	2603:1063:2000:1::12	2406:7400:c4:c47f:c...	TCP	503	[TCP Retransmission] 443 → 60179 [PSH, ACK] Seq=7546 Ack=232838 Win=49152 Len=429
17424	132.263235	2600:140f:3::685d:1...	2406:7400:c4:c47f:c...	TCP	158	[TCP Retransmission] 443 → 57458 [PSH, ACK] Seq=1192 Ack=5028 Win=75008 Len=84
17703	139.702410	2406:7400:c4:c47f:c...	2603:1063:2202:14::3	TCP	144	[TCP Retransmission] 50914 → 443 [PSH, ACK] Seq=5760 Ack=1099 Win=255 Len=70
17708	139.790747	2603:1063:5:122::36...	2406:7400:c4:c47f:c...	TCP	74	[TCP Retransmission] 443 → 52897 [FIN, ACK] Seq=5525 Ack=2676 Win=65280 Len=0
17710	139.790747	2603:1063:2001:1d2b...	2406:7400:c4:c47f:c...	TCP	74	[TCP Retransmission] 443 → 52899 [FIN, ACK] Seq=38411 Ack=4514 Win=64768 Len=0
17711	139.790747	2603:1063:5:122::36...	2406:7400:c4:c47f:c...	TCP	74	[TCP Retransmission] 443 → 52898 [FIN, ACK] Seq=5543 Ack=2663 Win=65280 Len=0
17713	139.790747	2603:1063:5:122::36...	2406:7400:c4:c47f:c...	TCP	74	[TCP Retransmission] 443 → 52895 [FIN, ACK] Seq=6772 Ack=4310 Win=65280 Len=0
17714	139.790747	2603:1063:5:122::36...	2406:7400:c4:c47f:c...	TCP	74	[TCP Retransmission] 443 → 52897 [FIN, ACK] Seq=5525 Ack=2676 Win=65280 Len=0
18173	148.092964	2406:7400:c4:c47f:c...	2603:1063:2000:1::12	TCP	1474	[TCP Retransmission] 60179 → 443 [PSH, ACK] Seq=398379 Ack=21501 Win=510 Len=1400
18657	161.238455	2603:1063:2000::12	2406:7400:c4:c47f:c...	TCP	1494	[TCP Retransmission] 443 → 49603 [PSH, ACK] Seq=7811 Ack=321...

THIS RE – TRANSMISSION SHOWS

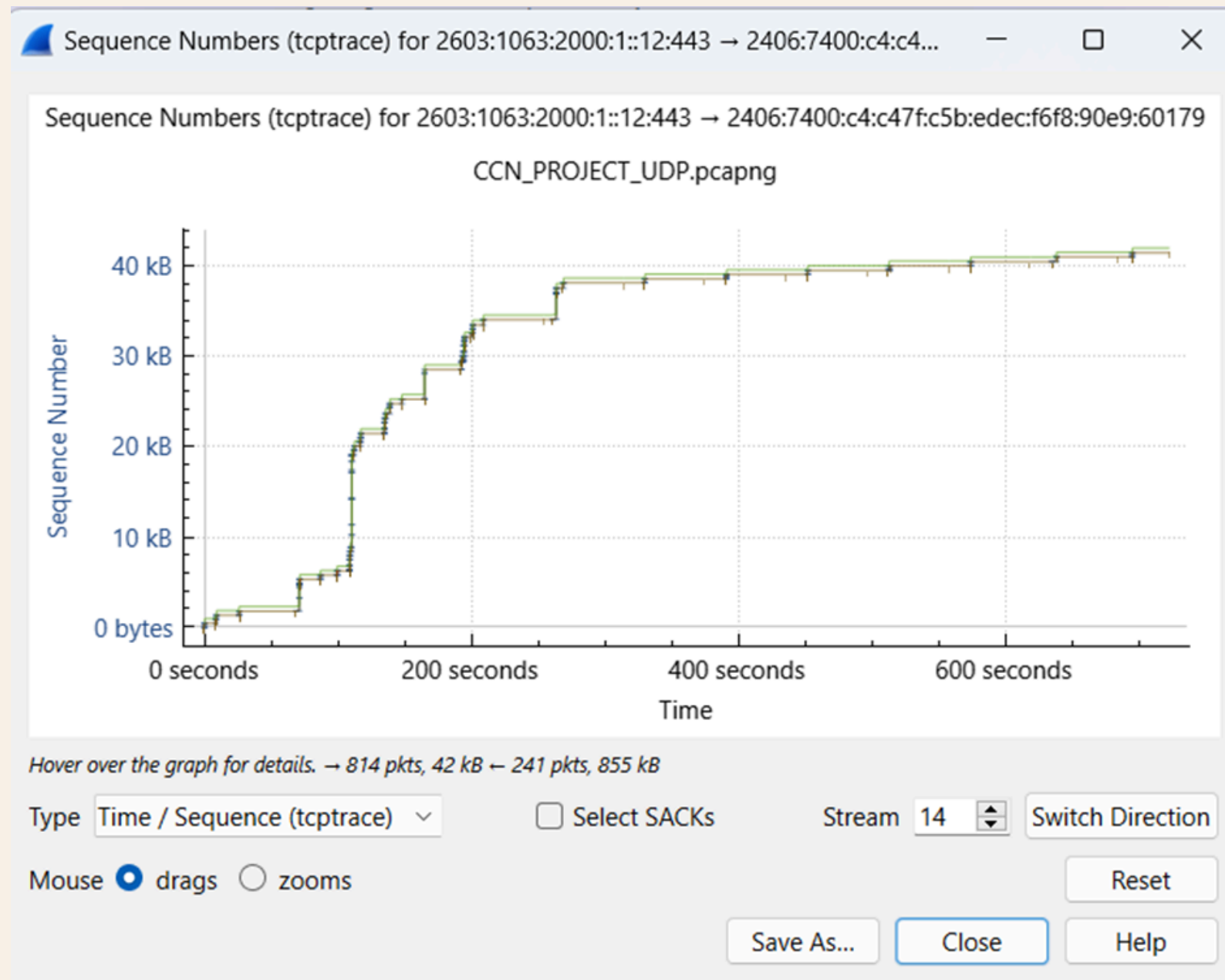
- Packet loss occurred
- TCP resends missing data

This proves TCP reliability: lost packets are detected and retransmitted

```
> Frame 535: Packet, 1474 bytes on wire (11792 bits), 1474 bytes captured (11792 bits) on interface \De
> Ethernet II, Src: Intel_af:c3:46 (34:7d:f6:af:c3:46), Dst: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b)
> Internet Protocol Version 6, Src: 2406:7400:c4:c47f:c5b:edec:f6f8:90e9, Dst: 2603:1063:2000:1::12
< Transmission Control Protocol, Src Port: 60179, Dst Port: 443, Seq: 6331, Ack: 1, Len: 1400
  Source Port: 60179
  Destination Port: 443
  [Stream index: 14]
  [Stream Packet Number: 4]
  > [Conversation completeness: Incomplete (28)]
  [TCP Segment Len: 1400]
  Sequence Number: 6331 (relative sequence number)
  Sequence Number (raw): 3236780000
  [Next Sequence Number: 7731 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 1562125421
  0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
  Window: 510
```

```
0030 00 00 00 00 00 12 eb 13 01 bb c0 ed 57 e0 5d 1c .....W..
0040 24 6d 50 18 01 fe 3b 80 00 00 df a3 1a 14 3b 8c Smp....;
0050 ba 38 e4 8c a3 0f 1c b4 3b 5e e3 40 08 3d d7 15 -8.....;^@=-
0060 10 17 21 2d fe f8 63 ca 13 60 44 47 fb f8 08 78 ..!...c. `DG...x
0070 72 cd d2 7c 5d 6c 09 ad 86 6d dd 40 6d d7 9c d8 r..]]l..m@m...
0080 e9 5d ca fb d3 0a 12 ab b7 a1 fb 4b c8 7e e8 ad .].....K~...
0090 03 9c e8 ff cb 36 0d 73 6d 00 fe 2e 8f eb 29 8d .....6.s m...
00a0 d0 0b e3 85 58 aa 3d 64 07 ce 7e df 73 8e 23 af ....X=d ~.s#
00b0 bd 18 7a 54 43 9a e1 a5 af 3f 97 c8 fb a9 63 8d ..zTC...?....c
00c0 4b f4 fa 0a ba e8 b4 6c ba dc 8e 77 d3 e6 eb 41 K.....] ...w...A
00d0 e6 30 5b 2b ff 59 40 47 ae 5c 45 ff ab 5b c7 a2 -0[+Y@G .\E..[
00e0 33 1a 5b 53 9f cc 73 eb 22 35 67 47 35 77 0b 40 3.[S...s "5g5W@
00f0 70 ab 43 3c c0 72 51 c0 08 b4 87 76 ed 68 3a fc p.C<-rQ. ...v-h:
0100 98 62 a5 0c 32 78 0d 0a 19 93 bf af 7e 66 5b 01 -b..2x...f[
0110 e2 a2 a0 9d 26 74 b6 d5 c6 d6 9c 42 53 cc ad 5c ...&t...BS.\
0120 54 d9 7f 7a 5a 95 20 25 80 f5 a0 30 c1 d7 14 ef T..zZ. % ...0....
0130 54 86 b9 75 16 d3 0c 9e ce 6f 54 2f 80 a4 f9 f4 T..u....oT/....
0140 b3 7f 7c 91 37 c1 40 b4 2a fa 5e c5 0f 17 72 8e ..|.7.@. *.^...r
0150 eb f0 b0 ef 95 5e f8 e3 60 3e 82 d4 8f 13 98 31 .....^..>.....1
0160 14 91 43 39 28 e0 d1 89 12 f2 df 6a 16 51 dd 19 ..C9( ...j-Q..
0170 44 86 82 e2 f3 5f ba 39 c3 1b 60 1f 03 48 69 03 D...._9 ...`..Hi
0180 0c a1 00 28 5b 78 7e 95 4f 72 d6 0b c3 b3 13 c1 ...([x~ Or.....
0190 a7 1e 8f 51 c1 64 50 42 2b ff 3f 8a 30 f0 01 06 ...Q.dPB +.?0...
```

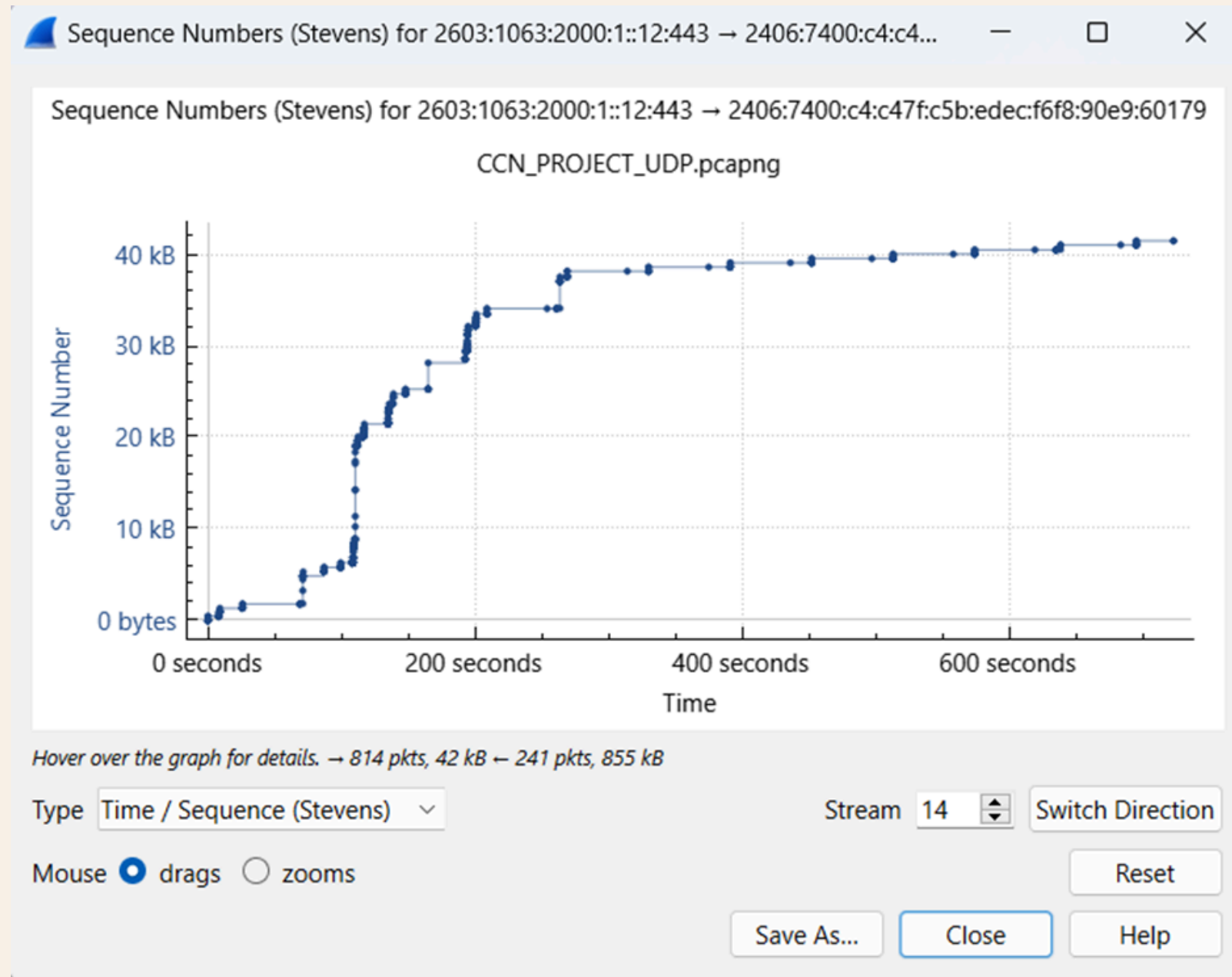
SEQUENCE NUMBERS (TCPTRACE):



```
Sequence Number (raw): 3236780000
[Next Sequence Number: 7731    (relative sequence number)]
Acknowledgment Number: 1    (relative ack number)
Acknowledgment number (raw): 1562125421
0101 .... = Header Length: 20 bytes (5)
> Flags: 0x018 (PSH, ACK)
Window: 510
[Calculated window size: 510]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x3b80 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
> [Timestamps]
> [SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (1400 bytes)
Retransmitted TCP segment data (1400 bytes)
```

Sequence numbers track packet order

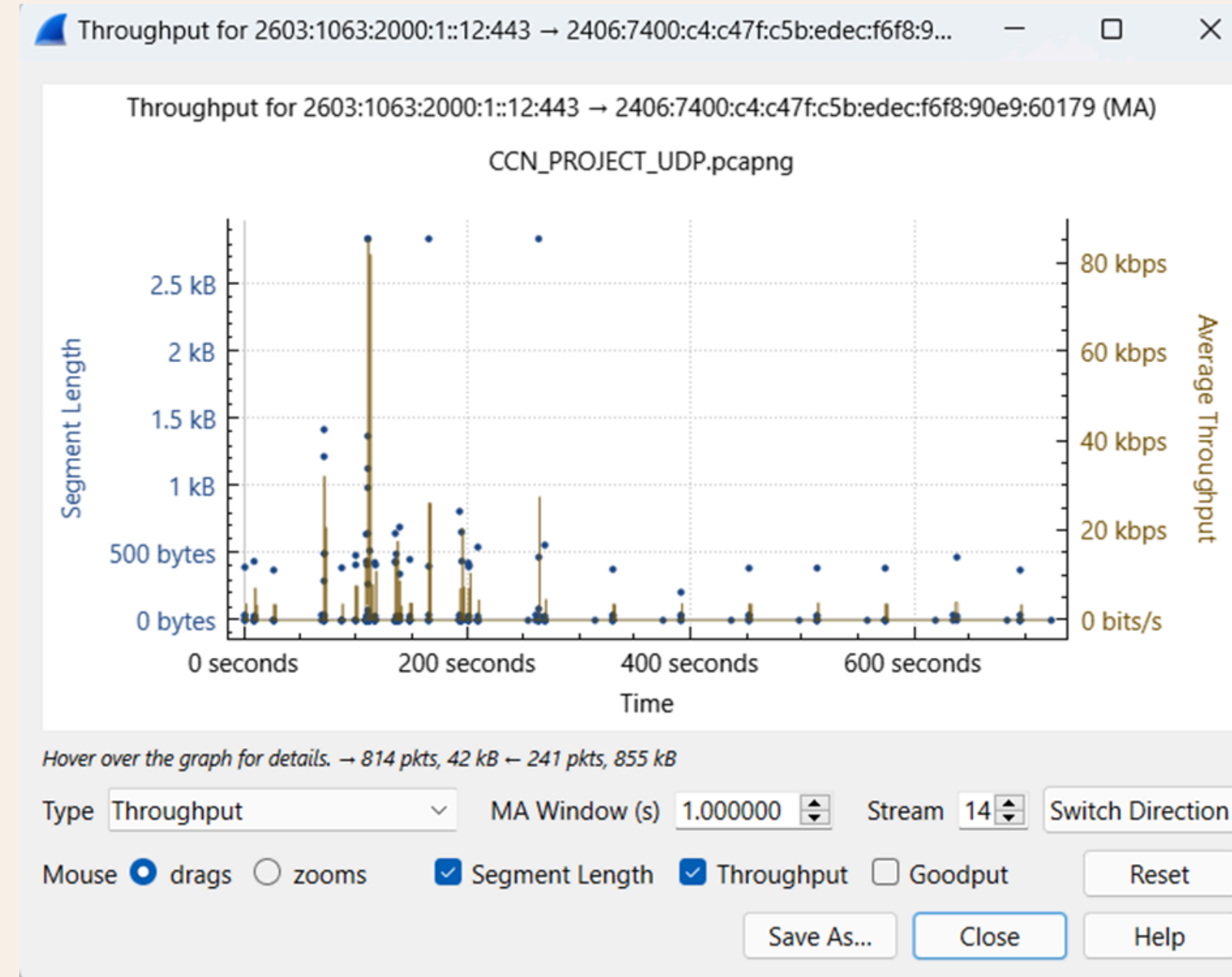
SEQUENCE NUMBERS (STREVEVS):



Helps detect:

- Loss
- Retransmission
- Flow control

THROUGHPUT:

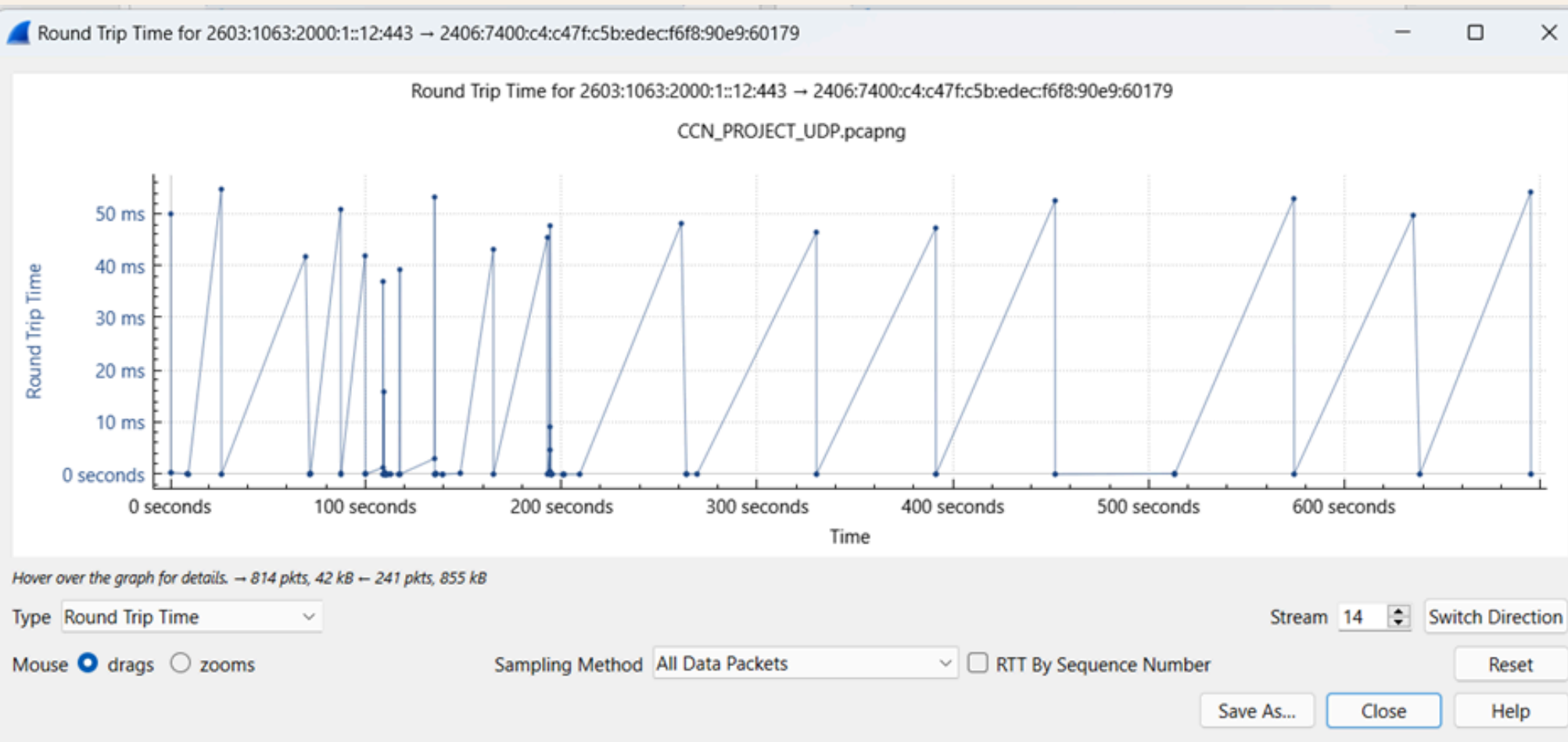


Shows how fast data is transmitted

Fluctuations indicate:

- Network congestion
- Variable speed

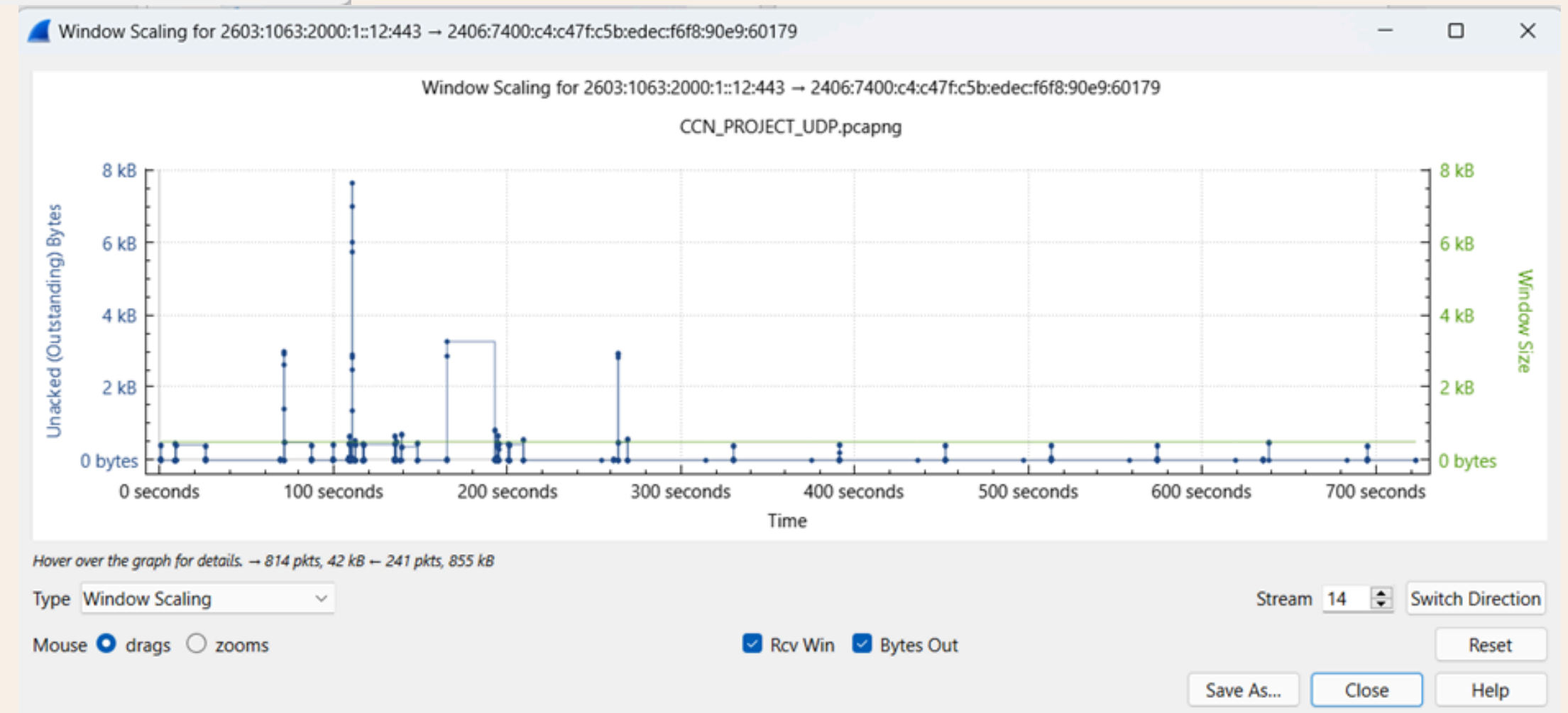
ROUND TIME GRAPH:



- Time taken for packet to go and return.
- RTT helps measure latency in TCP communication.
- Lower RTT means faster network.

WINDOW SCALING:

- Controls how much data can be sent before ACK
- Larger window means a higher performance
- So, Window scaling improves TCP efficiency in high-speed networks.



UDP ANALYSIS:

Packet flow:

udp							
No.	Time	Source	Destination	Protocol	Length	Info	
2331	11.763774	170.114.117.9	192.168.0.102	UDP	317	8801 → 60998 Len=271	
2332	11.763774	170.114.117.9	192.168.0.102	UDP	308	8801 → 60998 Len=262	
2333	11.763774	170.114.117.9	192.168.0.102	UDP	317	8801 → 54712 Len=271	
2334	11.763774	170.114.117.9	192.168.0.102	UDP	322	8801 → 54712 Len=276	
2335	11.763928	192.168.0.102	170.114.117.9	UDP	69	54712 → 8801 Len=27	
2336	11.763998	192.168.0.102	170.114.117.9	UDP	144	60998 → 8801 Len=102	
2337	11.770038	170.114.117.9	192.168.0.102	UDP	307	8801 → 60998 Len=261	
2338	11.770038	170.114.117.9	192.168.0.102	UDP	905	8801 → 54712 Len=859	
2339	11.770038	170.114.117.9	192.168.0.102	UDP	905	8801 → 54712 Len=859	
2340	11.771934	170.114.117.9	192.168.0.102	UDP	904	8801 → 54712 Len=858	
2341	11.782879	192.168.0.102	170.114.117.9	UDP	69	54712 → 8801 Len=27	
2342	11.791869	170.114.117.9	192.168.0.102	UDP	302	8801 → 60998 Len=256	
2343	11.791869	170.114.117.9	192.168.0.102	UDP	319	8801 → 54712 Len=273	
2344	11.792420	192.168.0.102	170.114.117.9	UDP	151	54712 → 8801 Len=109	
2345	11.796861	170.114.117.9	192.168.0.102	UDP	306	8801 → 60998 Len=260	
2346	11.827157	170.114.117.9	192.168.0.102	UDP	335	8801 → 54712 Len=289	

> Frame 20: Packet, 330 bytes on wire (2640 bits), 330 bytes captured (2640 bits) on interface \Device\NPF{...}

Ethernet II, Src: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b), Dst: Intel_af:c3:46 (34:7d:f6:af:c3:46)

> Destination: Intel_af:c3:46 (34:7d:f6:af:c3:46)

> Source: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b)

Type: 802.1Q Virtual LAN (0x8100)

[Stream index: 0]

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0

Internet Protocol Version 4, Src: 170.114.117.9, Dst: 192.168.0.102

User Datagram Protocol, Src Port: 8801, Dst Port: 60998

Data (284 bytes)

udp							
No.	Time	Source	Destination	Protocol	Length	Info	
65	5.089162	170.114.117.9	192.168.0.102	UDP	318	8801 → 60998 Len=272	
65	5.089162	170.114.117.9	192.168.0.102	UDP	905	8801 → 54712 Len=859	
62	5.089162	170.114.117.9	192.168.0.102	UDP	905	8801 → 54712 Len=859	
971	5.089162	170.114.117.9	192.168.0.102	UDP	904	8801 → 54712 Len=858	
972	5.089162	170.114.117.9	192.168.0.102	UDP	108	8801 → 54712 Len=62	
973	5.099359	192.168.0.102	170.114.117.9	UDP	69	54712 → 8801 Len=27	
974	5.115875	170.114.117.9	192.168.0.102	UDP	325	8801 → 60998 Len=279	
975	5.121504	170.114.117.9	192.168.0.102	UDP	325	8801 → 54712 Len=279	
976	5.152248	192.168.0.102	170.114.117.9	UDP	69	54712 → 8801 Len=27	
977	5.156492	170.114.117.9	192.168.0.102	UDP	321	8801 → 60998 Len=275	
978	5.182808	170.114.117.9	192.168.0.102	UDP	319	8801 → 54712 Len=273	
979	5.182808	170.114.117.9	192.168.0.102	UDP	904	8801 → 54712 Len=858	
980	5.182808	170.114.117.9	192.168.0.102	UDP	904	8801 → 54712 Len=858	
981	5.182808	170.114.117.9	192.168.0.102	UDP	903	8801 → 54712 Len=857	
982	5.182808	170.114.117.9	192.168.0.102	UDP	108	8801 → 54712 Len=62	
983	5.193354	192.168.0.102	170.114.117.9	UDP	76	54712 → 8801 Len=34	

> Frame 1: Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface \Device\NPF_{AE...}

Ethernet II, Src: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b), Dst: Intel_af:c3:46 (34:7d:f6:af:c3:46)

> Destination: Intel_af:c3:46 (34:7d:f6:af:c3:46)

> Source: TPLink_b4:dc:2b (5c:e9:31:b4:dc:2b)

Type: 802.1Q Virtual LAN (0x8100)

[Stream index: 0]

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0

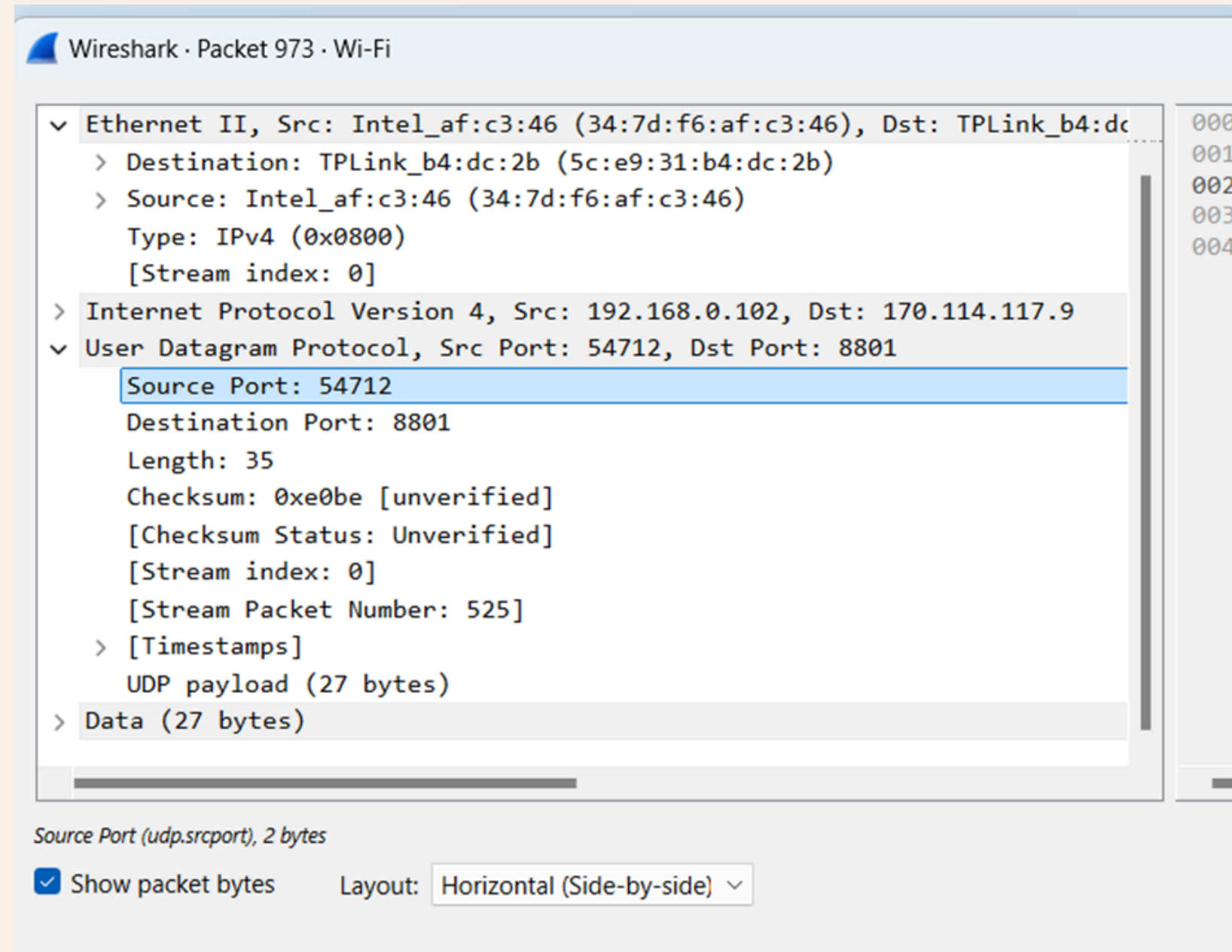
Internet Protocol Version 4, Src: 170.114.117.9, Dst: 192.168.0.102

User Datagram Protocol, Src Port: 8801, Dst Port: 54712

Data (27 bytes)

Packets flow without connection setup

USER DATAGRAM PROTOCOL:



Key UDP Features:

- No handshake
- No SYN / ACK anywhere
- No sequence numbers
- Unlike TCP, nothing tracking packets
- No retransmission
- No resend mechanism
- Small packets
- DNS packets are usually short

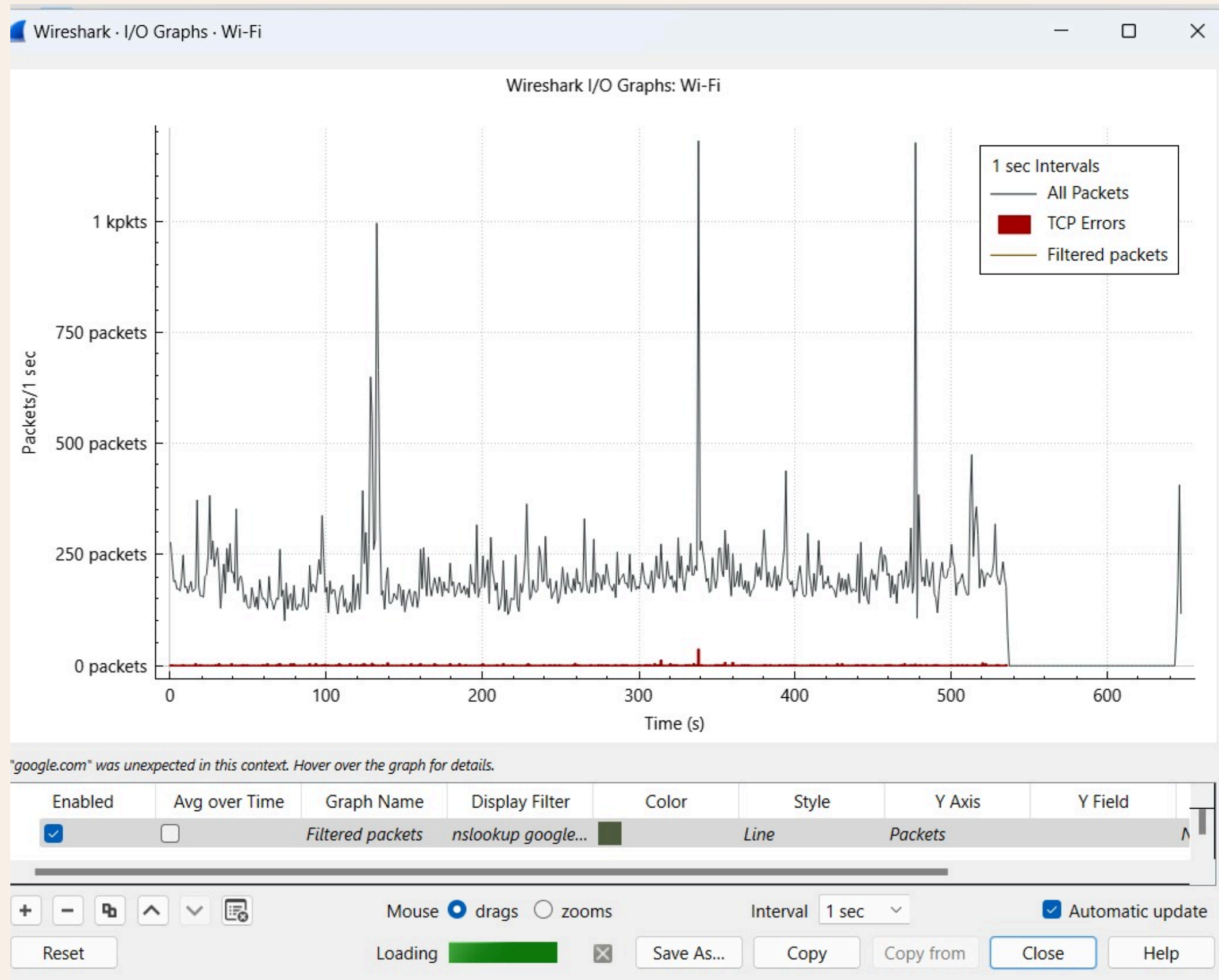
UDP is fast but unreliable — it sacrifices reliability for speed

DNS FILTER:

nslookup google.com						
No.	Time	Source	Destination	Protocol	Length	Info
42410	221.694378	2406:7400:c4:f8b7:a...	2406:7400:b0:b::2	DNS	97	Standard query 0x0072 A crl3.digicert.com
42411	221.694488	2406:7400:c4:f8b7:a...	2406:7400:b0:b::2	DNS	97	Standard query 0x1aa6 AAAA crl3.digicert.com
42413	221.697330	2406:7400:b0:b::2	2406:7400:c4:f8b7:a...	DNS	321	Standard query response 0x0072 A crl3.digicert.com CNAME crl.edge.digicert.com CNAME cdp1.digicert.com
42414	221.697330	2406:7400:b0:b::2	2406:7400:c4:f8b7:a...	DNS	333	Standard query response 0x1aa6 AAAA crl3.digicert.com CNAME crl.edge.digicert.com CNAME cdp1.digicert.com
42432	221.722019	2406:7400:c4:f8b7:a...	2406:7400:b0:b::2	DNS	94	Standard query 0x2530 A x2.c.lencr.org
42433	221.722089	2406:7400:c4:f8b7:a...	2406:7400:b0:b::2	DNS	94	Standard query 0x8795 AAAA x2.c.lencr.org
42434	221.724456	2406:7400:b0:b::2	2406:7400:c4:f8b7:a...	DNS	199	Standard query response 0x2530 A x2.c.lencr.org CNAME crl.root-x1.letsencrypt.org.edgekey.net CNAME e192961.dsc
42435	221.724456	2406:7400:b0:b::2	2406:7400:c4:f8b7:a...	DNS	239	Standard query response 0x8795 AAAA x2.c.lencr.org CNAME crl.root-x1.letsencrypt.org.edgekey.net CNAME e192961.dsc
42443	221.743001	2406:7400:c4:f8b7:a...	2406:7400:b0:b::2	DNS	94	Standard query 0x4eb8 A ye.c.lencr.org
42444	221.743131	2406:7400:c4:f8b7:a...	2406:7400:b0:b::2	DNS	94	Standard query 0xd3cd AAAA ye.c.lencr.org
42452	221.746082	2406:7400:b0:b::2	2406:7400:c4:f8b7:a...	DNS	204	Standard query response 0x4eb8 A ye.c.lencr.org CNAME e5.i.lencr.org.edgekey.net CNAME e192961.dsc
42453	221.746082	2406:7400:b0:b::2	2406:7400:c4:f8b7:a...	DNS	228	Standard query response 0xd3cd AAAA ye.c.lencr.org CNAME e5.i.lencr.org.edgekey.net CNAME e192961.dsc
42470	221.764974	2406:7400:c4:f8b7:a...	2406:7400:b0:b::2	DNS	103	Standard query 0xbb1b A ctldl.windowsupdate.com
42471	221.765139	2406:7400:c4:f8b7:a...	2406:7400:b0:b::2	DNS	103	Standard query 0xfd1a AAAA ctldl.windowsupdate.com
42472	221.766921	2406:7400:b0:b::2	2406:7400:c4:f8b7:a...	DNS	317	Standard query response 0xbb1b A ctldl.windowsupdate.com CNAME ctldl.windowsupdate.com.delivery.mi
42473	221.766921	2406:7400:b0:b::2	2406:7400:c4:f8b7:a...	DNS	341	Standard query response 0xfd1a AAAA ctldl.windowsupdate.com CNAME ctldl.windowsupdate.com.delivery.mi

- DNS queries are small and quick
- Uses UDP for speed

Overall traffic graph



OUR TEAM



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**THANK
YOU**